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DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Homen Dairy Farms

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 5511 W Sandy Mush RD | Merced | Merced | 95341-8404 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/26/2000

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X074-X080-X001-XXXX

**B. OPERATORS**

Homen, Frank

Operator name: Homen, Frank

Telephone no.: (209) 726-4728 Landline, (209) 564-1429 Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 5573 W Sandy Mush RD | Merced | CA | 95341 |

**This operator is responsible for paying permit fees.**

Homen, Tony

Operator name: Homen, Tony

Telephone no.: (209) 384-8133 Landline, (209) 564-1427 Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 5579 W Sandy Mush RD | Merced | CA | 95341 |

**C. OWNERS**

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Homen, Frank

| | | | | |
|---|---|---|---|---|
| Legal owner name: | Homen, Frank | Telephone no.: | (209) 726-4728 | (209) 564-1429 |
| | | | Landline | Cellular |
| 5573 W Sandy Mush RD | Merced | | CA | 95341 |
| Mailing Address Number and Street | City | | State | Zip Code |

**This owner is responsible for paying permit fees.**

Homen, Tony

| | | | | |
|---|---|---|---|---|
| Legal owner name: | Homen, Tony | Telephone no.: | (209) 384-8133 | (209) 564-1427 |
| | | | Landline | Cellular |
| 5579 W Sandy Mush RD | Merced | | CA | 95341 |
| Mailing Address Number and Street | City | | State | Zip Code |

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AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 0 | 0 | 0 |
| Number under roof | 2,060 | 270 | 1,050 | 930 | 390 | 300 |
| Maximum number | 2,100 | 300 | 1,100 | 1,000 | 400 | 330 |
| Average number | 2,060 | 270 | 1,050 | 930 | 390 | 300 |
| Avg live weight (lbs) | 1,400 | 1,400 | 900 | 650 | | |

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 80,355.07 *tons per reporting period*

Total nitrogen from manure: 988,606.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 692,024.84 *lbs per reporting period*

Total phosphorus from manure: 162,408.50 *lbs per reporting period*

Total potassium from manure: 405,336.37 *lbs per reporting period*

Total salt from manure: 1,032,037.50 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 121,320,000 *gallons*

Total nitrogen generated: 226,958.60 *lbs*

Total phosphorus generated: 24,489.49 *lbs*

Total potassium generated: 1,065,205.80 *lbs*

Total salt generated: 4,622,093.88 *lbs*

| | | |
|---|---|---|
| | 121,320,000 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 121,320,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Irrigation Wells | Ground water |
| IW 1 | Ground water |
| IW 2 | Ground water |
| IW 3 | Ground water |
| IW 4 | Ground water |

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| Source Description | Type |
|---|---|
| IW 5 | Ground water |
| IW 6 | Ground water |
| IW 7 | Ground water |
| IW 8 | Ground water |
| IW 9 | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

| Date | Material type / Description | Quantity | Reporting basis | Moisture (%) | N (%) | P (%) | K (%) | Salt (%) |
|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | Solid commercial fertilizer<br>*10-10-10-9S* | 92.25 *ton* | Dry-weight | 0.1 | 10.000000 | 10.000000 | 10.000000 | 0.000000 |

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 18,431.55 | 18,431.55 | 18,431.55 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total imports for all materials | 18,431.55 | 18,431.55 | 18,431.55 | 0.00 |

## G. NUTRIENT EXPORTS

| Date | Material type | Quantity | Reporting basis | Moisture (%) | Density (lbs/cu ft) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| 04/13/2023 | Corral solids | 7,799.26 *ton* | Dry-weight | 18.0 | | 8,900.00 | 3,400.00 | 18,600.00 | | 70.10 |
| 06/15/2023 | Corral solids | 2,827.33 *ton* | Dry-weight | 18.0 | | 8,900.00 | 3,400.00 | 18,600.00 | | 70.10 |
| 09/15/2023 | Separator solids | 3,600.00 *ton* | Dry-weight | 73.6 | | 16,300.00 | 4,000.00 | 22,800.00 | | 35.40 |

*No liquid nutrient exports entered.*

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Dry manure | 186,088.75 | 66,857.07 | 367,491.74 | 12,889,636.13 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total exports for all materials | 186,088.75 | 66,857.07 | 367,491.74 | 12,889,636.13 |

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APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 1 | 15 | 15 | 2 | both | X074-X080-X001-XXXX |
| Field 11 | 78 | 78 | 2 | both | X074-X080-X027-XXXX |
| Field 12 | 77 | 77 | 2 | both | X074-X080-X027-XXXX |
| Field 14 | 80 | 80 | 2 | process wastewater | X074-X080-X028-XXXX |
| Field 15 | 72 | 72 | 1 | process wastewater | X074-X080-X029-XXXX |
| Field 2 | 55 | 55 | 2 | both | X074-X080-X001-XXXX |
| Field 3 | 76 | 76 | 2 | process wastewater | X074-X080-X001-XXXX |
| Field 4 | 71 | 71 | 2 | both | X074-X080-X001-XXXX |
| Field 5 | 75 | 75 | 2 | both | X074-X080-X009-XXXX |
| Field 6 | 77 | 77 | 2 | both | X074-X080-X009-XXXX |
| Field 7 | 77 | 77 | 2 | manure | X074-X090-X024-XXXX |
| Field 8 | 72 | 72 | 1 | none | X074-X090-X024-XXXX |
| Field 9 | 77 | 77 | 1 | process wastewater | X074-X080-X004-XXXX |
| Totals for areas that were used for application | 830 | 830 | 22 | | |
| Totals for areas that were not used for application | 72 | 72 | 1 | | |
| Land application area totals | 902 | 902 | 23 | | |

## B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/01/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/20/2023 | 250.00 *ton* | As-is | | 62.4 | 5,800.00 | 1,000.00 | 7,500.00 | | 15.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 16.67 | 193.33 | 33.33 | 250.00 | 1,892.53 |

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## Field 1

### 05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/17/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/28/2023 | 300.00 *ton* | As-is | | 65.7 | 4,100.00 | 700.00 | 3,500.00 | | 5.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 20.00 | 164.00 | 28.00 | 140.00 | 740.88 |

## Field 11

Field name: Field 11

### 10/20/2022: Rye Silage

Crop: Rye Silage Acres planted: 78 Plant date: 10/20/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/19/2023 | 1,300.00 *ton* | As-is | | 63.0 | 11,300.00 | 1,700.00 | 16,000.00 | | 18.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 16.67 | 376.67 | 56.67 | 533.33 | 2,220.00 |

### 06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 06/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/27/2023 | 2,600.00 *ton* | As-is | | 66.4 | 4,800.00 | 800.00 | 5,900.00 | | 5.20 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 33.33 | 320.00 | 53.33 | 393.33 | 1,164.80 |

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## Field 12

Field name: Field 12

### 10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/22/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/19/2023 | 1,600.00 *ton* | As-is | | 76.1 | 5,600.00 | 1,000.00 | 7,500.00 | | 12.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 20.78 | 232.73 | 41.56 | 311.69 | 1,231.63 |

### 05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 05/24/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/05/2023 | 2,600.00 *ton* | As-is | | 66.5 | 4,100.00 | 700.00 | 4,500.00 | | 5.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 33.77 | 276.88 | 47.27 | 303.90 | 1,266.91 |

## Field 14

Field name: Field 14

### 11/01/2022: Oats, hay

Crop: Oats, hay Acres planted: 80 Plant date: 11/01/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 06/30/2023 | 400.00 *ton* | As-is | | 10.0 | 21,800.00 | 3,900.00 | 29,100.00 | | 12.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 6.00 | 244.80 | 41.40 | 149.40 | 0.00 |
| Total actual harvest content | 5.00 | 218.00 | 39.00 | 291.00 | 1,116.00 |

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## Field 14

### 07/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 07/04/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/27/2023 | 1,968.00 *ton* | As-is | | 72.2 | 3,600.00 | 800.00 | 4,200.00 | | 6.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 24.60 | 177.12 | 39.36 | 206.64 | 875.37 |

## Field 15

Field name: Field 15

### 09/20/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 72 Plant date: 09/20/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/26/2023 | 560.00 *ton* | As-is | | 10.9 | 32,800.00 | 2,700.00 | 30,000.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 10.00 | 583.00 | 62.00 | 357.00 | 0.00 |
| Total actual harvest content | 7.78 | 510.22 | 42.00 | 466.67 | 0.00 |

## Field 2

Field name: Field 2

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## Field 2

### 10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 10/03/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/19/2023 | 1,200.00 *ton* | As-is | | 67.9 | 5,200.00 | 900.00 | 7,700.00 | | 10.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 21.82 | 226.91 | 39.27 | 336.00 | 1,456.76 |

### 05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/25/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/05/2023 | 1,600.00 *ton* | As-is | | 69.8 | 3,800.00 | 700.00 | 4,300.00 | | 5.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 29.09 | 221.09 | 40.73 | 250.18 | 1,001.54 |

## Field 3

Field name: Field 3

### 09/10/2015: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 76 Plant date: 09/10/2015

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/28/2023 | 350.00 *ton* | As-is | | 6.9 | 20,600.00 | 2,700.00 | 23,000.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 6.00 | 349.80 | 37.20 | 214.20 | 0.00 |
| Total actual harvest content | 4.61 | 189.74 | 24.87 | 211.84 | 0.00 |

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## Field 3

### 09/30/2022: Rye Silage

Crop: Rye Silage Acres planted: 76 Plant date: 09/30/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/19/2023 | 240.00 *ton* | As-is | | 29.6 | 22,400.00 | 3,000.00 | 28,600.00 | | 11.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 3.16 | 141.47 | 18.95 | 180.63 | 515.77 |

## Field 4

Field name: Field 4

### 10/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 71 Plant date: 10/02/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/12/2023 | 750.00 *ton* | As-is | | 70.3 | 8,900.00 | 1,600.00 | 10,100.00 | | 13.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 10.56 | 188.03 | 33.80 | 213.38 | 865.90 |

### 05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 71 Plant date: 05/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/28/2023 | 2,200.00 *ton* | As-is | | 67.9 | 4,300.00 | 800.00 | 4,600.00 | | 5.50 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 30.99 | 266.48 | 49.58 | 285.07 | 1,094.11 |

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## Field 5

Field name: Field 5

### 10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 10/10/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/10/2023 | 1,010.00 *ton* | As-is | | 70.3 | 8,900.00 | 1,600.00 | 10,100.00 | | 13.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 13.47 | 239.71 | 43.09 | 272.03 | 1,103.89 |

### 06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/26/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/13/2023 | 2,200.00 *ton* | As-is | | 71.3 | 3,700.00 | 800.00 | 4,200.00 | | 5.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 29.33 | 217.07 | 46.93 | 246.40 | 993.40 |

## Field 6

Field name: Field 6

### 10/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/02/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/12/2023 | 900.00 *ton* | As-is | | 68.5 | 7,600.00 | 1,500.00 | 9,500.00 | | 13.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 11.69 | 177.66 | 35.06 | 222.08 | 979.36 |

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## Field 6

### 05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 05/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/29/2023 | 2,400.00 *ton* | As-is | | 69.4 | 3,900.00 | 600.00 | 4,400.00 | | 5.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 31.17 | 243.12 | 37.40 | 274.29 | 1,106.37 |

## Field 7

Field name: Field 7

### 10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/10/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/19/2023 | 1,300.00 *ton* | As-is | | 76.1 | 5,800.00 | 100.00 | 8,100.00 | | 12.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 17.00 | 231.20 | 39.10 | 141.10 | 0.00 |
| Total actual harvest content | 16.88 | 195.84 | 3.38 | 273.51 | 976.49 |

### 05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 05/22/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/04/2023 | 2,200.00 *ton* | As-is | | 71.2 | 3,300.00 | 700.00 | 3,900.00 | | 5.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 274.40 | 44.80 | 184.80 | 0.00 |
| Total actual harvest content | 28.57 | 188.57 | 40.00 | 222.86 | 888.69 |

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## Field 8

Field name: Field 8

### 10/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 72 Plant date: 10/01/2020

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/30/2023 | 600.00 *ton* | As-is | | 9.2 | 38,400.00 | 2,800.00 | 26,000.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 10.00 | 583.00 | 62.00 | 357.00 | 0.00 |
| Total actual harvest content | 8.33 | 640.00 | 46.67 | 433.33 | 0.00 |

## Field 9

Field name: Field 9

### 01/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 77 Plant date: 01/01/2020

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/30/2023 | 700.00 *ton* | As-is | | 9.6 | 35,700.00 | 2,700.00 | 27,000.00 | | 0.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 10.00 | 583.00 | 62.00 | 357.00 | 0.00 |
| Total actual harvest content | 9.09 | 649.09 | 49.09 | 490.91 | 0.00 |

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NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Field 1 - 10/01/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough  Plant date: 10/01/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/13/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Compost | 187.33 | 17.89 | 47.36 | 0.00 | 83.00 *ton* |
| Application event totals | | 187.33 | 17.89 | 47.36 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 02/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 56.18 | 6.01 | 354.50 | 1,472.06 | 540,000.00 *gal* |
| Irrigation Wells | Ground water | 16.66 | 0.00 | 0.00 | 1,249.39 | 2,023,200.00 *gal* |
| Application event totals | | 72.84 | 6.01 | 354.50 | 2,721.44 | |

### Field 1 - 05/17/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage  Plant date: 05/17/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 21.52 | 0.00 | 0.00 | 1,613.79 | 2,613,300.00 *gal* |
| Application event totals | | 21.52 | 0.00 | 0.00 | 1,613.79 | |

| | | | | |
|---|---|---|---|---|
| 06/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.43 | 0.00 | 0.00 | 1,457.62 | 2,360,400.00 *gal* |
| Application event totals | | 19.43 | 0.00 | 0.00 | 1,457.62 | |

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## Field 1 - 05/17/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 135.43 | 10.37 | 177.77 | 953.33 | 504,000.00 *gal* |
| Irrigation Wells | Ground water | 16.66 | 0.00 | 0.00 | 1,249.39 | 2,023,200.00 *gal* |
| Application event totals | | 152.09 | 10.37 | 177.77 | 2,202.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 14.58 | 0.00 | 0.00 | 1,093.21 | 1,770,300.00 *gal* |
| Application event totals | | 14.58 | 0.00 | 0.00 | 1,093.21 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 13.88 | 0.00 | 0.00 | 1,041.16 | 1,686,000.00 *gal* |
| Application event totals | | 13.88 | 0.00 | 0.00 | 1,041.16 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.35 | 0.00 | 0.00 | 1,301.44 | 2,107,500.00 *gal* |
| Application event totals | | 17.35 | 0.00 | 0.00 | 1,301.44 | |

## Field 11 - 10/20/2022: Rye Silage

Field name: Field 11

Crop: Rye Silage

Plant date: 10/20/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/19/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Compost | 176.66 | 16.87 | 44.66 | 0.00 | 407.00 *ton* |
| Application event totals | | 176.66 | 16.87 | 44.66 | 0.00 | |

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## Field 11 - 10/20/2022: Rye Silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 74.18 | 7.93 | 468.12 | 1,943.87 | 3,708,000.00 *gal* |
| Irrigation Wells | Ground water | 11.94 | 0.00 | 0.00 | 887.60 | 7,341,840.00 *gal* |
| Application event totals | | 86.12 | 7.93 | 468.12 | 2,831.47 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 64.10 | 6.86 | 404.49 | 1,679.66 | 3,204,000.00 *gal* |
| Irrigation Wells | Ground water | 10.32 | 0.00 | 0.00 | 766.95 | 6,343,920.00 *gal* |
| Application event totals | | 74.42 | 6.86 | 404.49 | 2,446.61 | |

## Field 11 - 06/16/2023: Corn, silage

Field name: Field 11

Crop: Corn, silage

Plant date: 06/16/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/02/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 72.69 | 10.29 | 30.88 | 0.00 | 980.00 *ton* |
| Application event totals | | 72.69 | 10.29 | 30.88 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 20.40 | 0.00 | 0.00 | 1,516.67 | 12,545,280.00 *gal* |
| Application event totals | | 20.40 | 0.00 | 0.00 | 1,516.67 | |

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## Field 11 - 06/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 92.44 | 12.08 | 653.22 | 2,773.11 | 5,760,000.00 *gal* |
| Irrigation Wells | Ground water | 18.55 | 0.00 | 0.00 | 1,378.79 | 11,404,800.00 *gal* |
| Application event totals | | 110.98 | 12.08 | 653.22 | 4,151.90 | |

| | | | | |
|---|---|---|---|---|
| 07/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.62 | 0.00 | 0.00 | 1,309.85 | 10,834,560.00 *gal* |
| Application event totals | | 17.62 | 0.00 | 0.00 | 1,309.85 | |

| | | | | |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 18.31 | 0.00 | 0.00 | 1,361.55 | 11,262,240.00 *gal* |
| Application event totals | | 18.31 | 0.00 | 0.00 | 1,361.55 | |

| | | | | |
|---|---|---|---|---|
| 08/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 95.90 | 12.53 | 677.72 | 2,877.10 | 5,976,000.00 *gal* |
| Irrigation Wells | Ground water | 19.24 | 0.00 | 0.00 | 1,430.49 | 11,832,480.00 *gal* |
| Application event totals | | 115.15 | 12.53 | 677.72 | 4,307.59 | |

| | | | | |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.39 | 0.00 | 0.00 | 1,292.61 | 10,692,000.00 *gal* |
| Application event totals | | 17.39 | 0.00 | 0.00 | 1,292.61 | |

## Field 12 - 10/22/2022: Oats, silage-soft dough

Field name: Field 12

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

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## Field 12 - 10/22/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/17/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 97.04 | 10.21 | 26.17 | 0.00 | 1,078.00 *ton* |
| Application event totals | | 97.04 | 10.21 | 26.17 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/19/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 87.29 | 11.63 | 471.23 | 2,039.42 | 3,564,000.00 *gal* |
| Irrigation Wells | Ground water | 11.62 | 0.00 | 0.00 | 864.21 | 7,056,720.00 *gal* |
| Application event totals | | 98.92 | 11.63 | 471.23 | 2,903.63 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 67.85 | 7.26 | 428.16 | 1,777.94 | 3,348,000.00 *gal* |
| Irrigation Wells | Ground water | 10.92 | 0.00 | 0.00 | 811.83 | 6,629,040.00 *gal* |
| Application event totals | | 78.77 | 7.26 | 428.16 | 2,589.77 | |

## Field 12 - 05/24/2023: Corn, silage

Field name: Field 12

Crop: Corn, silage

Plant date: 05/24/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.14 | 0.00 | 0.00 | 1,422.88 | 11,618,640.00 *gal* |
| Application event totals | | 19.14 | 0.00 | 0.00 | 1,422.88 | |

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Field 12 - 05/24/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.03 | 0.00 | 0.00 | 1,265.76 | 10,335,600.00 *gal* |
| Application event totals | | 17.03 | 0.00 | 0.00 | 1,265.76 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 173.37 | 13.28 | 227.57 | 1,220.41 | 3,312,000.00 *gal* |
| Irrigation Wells | Ground water | 15.73 | 0.00 | 0.00 | 1,169.73 | 9,551,520.00 *gal* |
| Application event totals | | 189.10 | 13.28 | 227.57 | 2,390.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 16.44 | 0.00 | 0.00 | 1,222.11 | 9,979,200.00 *gal* |
| Application event totals | | 16.44 | 0.00 | 0.00 | 1,222.11 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/20/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 74.32 | 9.71 | 525.23 | 2,229.74 | 4,572,000.00 *gal* |
| Irrigation Wells | Ground water | 14.91 | 0.00 | 0.00 | 1,108.63 | 9,052,560.00 *gal* |
| Application event totals | | 89.24 | 9.71 | 525.23 | 3,338.37 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 13.97 | 0.00 | 0.00 | 1,038.79 | 8,482,320.00 *gal* |
| Application event totals | | 13.97 | 0.00 | 0.00 | 1,038.79 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 15.15 | 0.00 | 0.00 | 1,126.09 | 9,195,120.00 *gal* |
| Application event totals | | 15.15 | 0.00 | 0.00 | 1,126.09 | |

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## Field 14 - 11/01/2022: Oats, hay

Field name: Field 14

Crop: Oats, hay

Plant date: 11/01/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 70.22 | 7.51 | 443.12 | 1,840.07 | 3,600,000.00 *gal* |
| Irrigation Wells | Ground water | 11.74 | 0.00 | 0.00 | 876.23 | 7,500,000.00 *gal* |
| Application event totals | | 81.96 | 7.51 | 443.12 | 2,716.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 163.24 | 12.50 | 214.27 | 1,149.11 | 3,240,000.00 *gal* |
| Irrigation Wells | Ground water | 10.56 | 0.00 | 0.00 | 788.60 | 6,750,000.00 *gal* |
| Application event totals | | 173.80 | 12.50 | 214.27 | 1,937.71 | |

## Field 14 - 07/04/2023: Corn, silage

Field name: Field 14

Crop: Corn, silage

Plant date: 07/04/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 21.36 | 0.00 | 0.00 | 1,594.73 | 13,650,000.00 *gal* |
| Application event totals | | 21.36 | 0.00 | 0.00 | 1,594.73 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 63.09 | 8.24 | 445.82 | 1,892.65 | 4,032,000.00 *gal* |
| Irrigation Wells | Ground water | 19.01 | 0.00 | 0.00 | 1,419.48 | 12,150,000.00 *gal* |
| Application event totals | | 82.10 | 8.24 | 445.82 | 3,312.13 | |

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## Field 14 - 07/04/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.84 | 0.00 | 0.00 | 1,331.86 | 11,400,000.00 *gal* |
| Application event totals | | 17.84 | 0.00 | 0.00 | 1,331.86 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.60 | 0.00 | 0.00 | 1,463.30 | 12,525,000.00 *gal* |
| Application event totals | | 19.60 | 0.00 | 0.00 | 1,463.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 55.20 | 7.21 | 390.10 | 1,656.07 | 3,528,000.00 *gal* |
| Irrigation Wells | Ground water | 16.66 | 0.00 | 0.00 | 1,244.24 | 10,650,000.00 *gal* |
| Application event totals | | 71.87 | 7.21 | 390.10 | 2,900.30 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 16.08 | 0.00 | 0.00 | 1,200.43 | 10,275,000.00 *gal* |
| Application event totals | | 16.08 | 0.00 | 0.00 | 1,200.43 | |

## Field 15 - 09/20/2022: Alfalfa, hay

Field name: Field 15

Crop: Alfalfa, hay

Plant date: 09/20/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

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Field 15 - 09/20/2022: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 261.99 | 20.07 | 343.90 | 1,844.25 | 4,680,000.00 *gal* |
| Irrigation Wells | Ground water | 21.12 | 0.00 | 0.00 | 1,577.21 | 12,150,000.00 *gal* |
| Application event totals | | 283.11 | 20.07 | 343.90 | 3,421.45 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 10-10-10-9S | Solid commercial fertilizer | 50.00 | 50.00 | 50.00 | 0.00 | |
| Irrigation Wells | Ground water | 19.30 | 0.00 | 0.00 | 1,440.90 | 11,100,000.00 *gal* |
| Application event totals | | 69.30 | 50.00 | 50.00 | 1,440.90 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 18.78 | 0.00 | 0.00 | 1,401.96 | 10,800,000.00 *gal* |
| Application event totals | | 18.78 | 0.00 | 0.00 | 1,401.96 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.34 | 0.00 | 0.00 | 1,294.87 | 9,975,000.00 *gal* |
| Application event totals | | 17.34 | 0.00 | 0.00 | 1,294.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 16.69 | 0.00 | 0.00 | 1,246.19 | 9,600,000.00 *gal* |
| Application event totals | | 16.69 | 0.00 | 0.00 | 1,246.19 | |

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## Field 15 - 09/20/2022: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 88.25 | 11.53 | 623.62 | 2,647.45 | 5,076,000.00 *gal* |
| Irrigation Wells | Ground water | 18.39 | 0.00 | 0.00 | 1,372.75 | 10,575,000.00 *gal* |
| Application event totals | | 106.63 | 11.53 | 623.62 | 4,020.20 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 16.04 | 0.00 | 0.00 | 1,197.51 | 9,225,000.00 *gal* |
| Application event totals | | 16.04 | 0.00 | 0.00 | 1,197.51 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 15.65 | 0.00 | 0.00 | 1,168.30 | 9,000,000.00 *gal* |
| Application event totals | | 15.65 | 0.00 | 0.00 | 1,168.30 | |

## Field 2 - 10/03/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/15/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 99.99 | 13.32 | 539.77 | 2,336.07 | 2,916,000.00 *gal* |
| Application event totals | | 99.99 | 13.32 | 539.77 | 2,336.07 | |

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## Field 2 - 10/03/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 98.06 | 10.49 | 618.76 | 2,569.41 | 3,456,000.00 *gal* |
| Irrigation Wells | Ground water | 25.48 | 0.00 | 0.00 | 1,890.11 | 11,122,560.00 *gal* |
| Application event totals | | 123.54 | 10.49 | 618.76 | 4,459.52 | |

## Field 2 - 05/25/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/25/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/29/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 62.69 | 8.88 | 26.63 | 0.00 | 596.00 *ton* |
| Application event totals | | 62.69 | 8.88 | 26.63 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 05/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 21.77 | 0.00 | 0.00 | 1,614.47 | 9,500,520.00 *gal* |
| Application event totals | | 21.77 | 0.00 | 0.00 | 1,614.47 | |

| | | | | |
|---|---|---|---|---|
| 06/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.91 | 0.00 | 0.00 | 1,476.65 | 8,689,500.00 *gal* |
| Application event totals | | 19.91 | 0.00 | 0.00 | 1,476.65 | |

| | | | | |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.11 | 0.00 | 0.00 | 1,417.58 | 8,341,920.00 *gal* |
| Application event totals | | 19.11 | 0.00 | 0.00 | 1,417.58 | |

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## Field 2 - 05/25/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 57.35 | 7.49 | 405.29 | 1,720.59 | 2,520,000.00 *gal* |
| Irrigation Wells | Ground water | 18.58 | 0.00 | 0.00 | 1,378.20 | 8,110,200.00 *gal* |
| Application event totals | | 75.93 | 7.49 | 405.29 | 3,098.79 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.52 | 0.00 | 0.00 | 1,299.45 | 7,646,760.00 *gal* |
| Application event totals | | 17.52 | 0.00 | 0.00 | 1,299.45 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 19.64 | 0.00 | 0.00 | 1,456.96 | 8,573,640.00 *gal* |
| Application event totals | | 19.64 | 0.00 | 0.00 | 1,456.96 | |

## Field 3 - 09/10/2015: Alfalfa, hay

Field name: Field 3

Crop: Alfalfa, hay

Plant date: 09/10/2015

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 17.29 | 0.00 | 0.00 | 1,282.35 | 10,427,400.00 *gal* |
| Application event totals | | 17.29 | 0.00 | 0.00 | 1,282.35 | |

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Field 3 - 09/10/2015: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 10-10-10-9S | Solid commercial fertilizer | 50.00 | 50.00 | 50.00 | 0.00 | |
| Irrigation Wells | Ground water | 15.75 | 0.00 | 0.00 | 1,168.36 | 9,500,520.00 *gal* |
| Application event totals | | 65.75 | 50.00 | 50.00 | 1,168.36 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 15.56 | 0.00 | 0.00 | 1,154.12 | 9,384,660.00 *gal* |
| Application event totals | | 15.56 | 0.00 | 0.00 | 1,154.12 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 15.18 | 0.00 | 0.00 | 1,125.62 | 9,152,940.00 *gal* |
| Application event totals | | 15.18 | 0.00 | 0.00 | 1,125.62 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 16.52 | 0.00 | 0.00 | 1,225.36 | 9,963,960.00 *gal* |
| Application event totals | | 16.52 | 0.00 | 0.00 | 1,225.36 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 14.60 | 0.00 | 0.00 | 1,082.87 | 8,805,360.00 *gal* |
| Application event totals | | 14.60 | 0.00 | 0.00 | 1,082.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/29/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 53.36 | 6.97 | 377.11 | 1,600.92 | 3,240,000.00 *gal* |
| Irrigation Wells | Ground water | 17.29 | 0.00 | 0.00 | 1,282.35 | 10,427,400.00 *gal* |
| Application event totals | | 70.65 | 6.97 | 377.11 | 2,883.27 | |

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## Field 3 - 09/10/2015: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 15.75 | 0.00 | 0.00 | 1,168.36 | 9,500,520.00 *gal* |
| Application event totals | | 15.75 | 0.00 | 0.00 | 1,168.36 | |

## Field 3 - 09/30/2022: Rye Silage

Field name: Field 3

Crop: Rye Silage

Plant date: 09/30/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/18/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 72.36 | 9.64 | 390.63 | 1,690.57 | 2,916,000.00 *gal* |
| Irrigation Wells | Ground water | 22.86 | 0.00 | 0.00 | 1,695.55 | 13,787,340.00 *gal* |
| Application event totals | | 95.22 | 9.64 | 390.63 | 3,386.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 65.79 | 7.04 | 415.13 | 1,723.86 | 3,204,000.00 *gal* |
| Irrigation Wells | Ground water | 19.21 | 0.00 | 0.00 | 1,424.83 | 11,586,000.00 *gal* |
| Application event totals | | 85.00 | 7.04 | 415.13 | 3,148.69 | |

## Field 4 - 10/02/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 10/02/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

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## Field 4 - 10/02/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/05/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 87.98 | 11.72 | 474.92 | 2,055.38 | 3,312,000.00 *gal* |
| Irrigation Wells | Ground water | 3.28 | 0.00 | 0.00 | 262.63 | 5,078,400.00 *gal* |
| Irrigation Wells | Ground water | 12.16 | 0.00 | 0.00 | 908.31 | 6,900,000.00 *gal* |
| Application event totals | | 103.42 | 11.72 | 474.92 | 3,226.32 | |

| | | | | |
|---|---|---|---|---|
| 01/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 79.12 | 8.46 | 499.29 | 2,073.32 | 3,600,000.00 *gal* |
| Irrigation Wells | Ground water | 3.56 | 0.00 | 0.00 | 285.47 | 5,520,000.00 *gal* |
| Irrigation Wells | Ground water | 13.22 | 0.00 | 0.00 | 987.30 | 7,500,000.00 *gal* |
| Application event totals | | 95.91 | 8.46 | 499.29 | 3,346.09 | |

## Field 4 - 05/16/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/16/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/18/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 65.19 | 9.23 | 27.69 | 0.00 | 800.00 *ton* |
| Application event totals | | 65.19 | 9.23 | 27.69 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 3.88 | 0.00 | 0.00 | 311.16 | 6,016,800.00 *gal* |
| Irrigation Wells | Ground water | 14.41 | 0.00 | 0.00 | 1,076.15 | 8,175,000.00 *gal* |
| Application event totals | | 18.30 | 0.00 | 0.00 | 1,387.31 | |

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Field 4 - 05/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 100.14 | 7.67 | 131.45 | 704.93 | 1,764,000.00 *gal* |
| Irrigation Wells | Ground water | 3.46 | 0.00 | 0.00 | 276.91 | 5,354,400.00 *gal* |
| Irrigation Wells | Ground water | 12.83 | 0.00 | 0.00 | 957.68 | 7,275,000.00 *gal* |
| Application event totals | | 116.42 | 7.67 | 131.45 | 1,939.51 | |

| | | | | |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 3.21 | 0.00 | 0.00 | 256.92 | 4,968,000.00 *gal* |
| Irrigation Wells | Ground water | 11.90 | 0.00 | 0.00 | 888.57 | 6,750,000.00 *gal* |
| Application event totals | | 15.11 | 0.00 | 0.00 | 1,145.49 | |

| | | | | |
|---|---|---|---|---|
| 07/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 53.95 | 7.05 | 381.24 | 1,618.46 | 3,060,000.00 *gal* |
| Irrigation Wells | Ground water | 3.03 | 0.00 | 0.00 | 242.65 | 4,692,000.00 *gal* |
| Irrigation Wells | Ground water | 11.24 | 0.00 | 0.00 | 839.20 | 6,375,000.00 *gal* |
| Application event totals | | 68.22 | 7.05 | 381.24 | 2,700.31 | |

| | | | | |
|---|---|---|---|---|
| 07/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 3.17 | 0.00 | 0.00 | 254.07 | 4,912,800.00 *gal* |
| Irrigation Wells | Ground water | 11.77 | 0.00 | 0.00 | 878.69 | 6,675,000.00 *gal* |
| Application event totals | | 14.94 | 0.00 | 0.00 | 1,132.76 | |

| | | | | |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 3.31 | 0.00 | 0.00 | 265.49 | 5,133,600.00 *gal* |
| Irrigation Wells | Ground water | 12.30 | 0.00 | 0.00 | 918.19 | 6,975,000.00 *gal* |
| Application event totals | | 15.61 | 0.00 | 0.00 | 1,183.67 | |

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## Field 5 - 10/10/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/28/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 89.62 | 11.94 | 483.80 | 2,093.81 | 3,564,000.00 *gal* |
| Irrigation Wells | Ground water | 10.31 | 0.00 | 0.00 | 775.03 | 6,219,180.00 *gal* |
| Irrigation Wells | Ground water | 12.39 | 0.00 | 0.00 | 925.29 | 7,425,000.00 *gal* |
| Application event totals | | 112.32 | 11.94 | 483.80 | 3,794.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 68.16 | 7.29 | 430.12 | 1,786.10 | 3,276,000.00 *gal* |
| Irrigation Wells | Ground water | 9.48 | 0.00 | 0.00 | 712.40 | 5,716,620.00 *gal* |
| Irrigation Wells | Ground water | 11.39 | 0.00 | 0.00 | 850.52 | 6,825,000.00 *gal* |
| Application event totals | | 89.03 | 7.29 | 430.12 | 3,349.02 | |

## Field 5 - 06/26/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/26/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/13/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 64.64 | 9.15 | 27.46 | 0.00 | 838.00 *ton* |
| Application event totals | | 64.64 | 9.15 | 27.46 | 0.00 | |

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Field 5 - 06/26/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 8.75 | 0.00 | 0.00 | 657.60 | 5,276,880.00 *gal* |
| Irrigation Wells | Ground water | 10.51 | 0.00 | 0.00 | 785.10 | 6,300,000.00 *gal* |
| Application event totals | | 19.26 | 0.00 | 0.00 | 1,442.70 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 45.66 | 5.97 | 322.69 | 1,369.92 | 2,736,000.00 *gal* |
| Irrigation Wells | Ground water | 7.92 | 0.00 | 0.00 | 594.97 | 4,774,320.00 *gal* |
| Irrigation Wells | Ground water | 9.51 | 0.00 | 0.00 | 710.33 | 5,700,000.00 *gal* |
| Application event totals | | 63.09 | 5.97 | 322.69 | 2,675.21 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 8.64 | 0.00 | 0.00 | 649.77 | 5,214,060.00 *gal* |
| Irrigation Wells | Ground water | 10.39 | 0.00 | 0.00 | 775.75 | 6,225,000.00 *gal* |
| Application event totals | | 19.03 | 0.00 | 0.00 | 1,425.52 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 43.86 | 5.73 | 309.95 | 1,315.84 | 2,628,000.00 *gal* |
| Irrigation Wells | Ground water | 7.60 | 0.00 | 0.00 | 571.48 | 4,585,860.00 *gal* |
| Irrigation Wells | Ground water | 9.14 | 0.00 | 0.00 | 682.29 | 5,475,000.00 *gal* |
| Application event totals | | 60.60 | 5.73 | 309.95 | 2,569.61 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 7.19 | 0.00 | 0.00 | 540.17 | 4,334,580.00 *gal* |
| Irrigation Wells | Ground water | 8.64 | 0.00 | 0.00 | 644.90 | 5,175,000.00 *gal* |
| Application event totals | | 15.82 | 0.00 | 0.00 | 1,185.07 | |

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## Field 5 - 06/26/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 6.67 | 0.00 | 0.00 | 501.03 | 4,020,480.00 *gal* |
| Irrigation Wells | Ground water | 8.01 | 0.00 | 0.00 | 598.17 | 4,800,000.00 *gal* |
| Application event totals | | 14.68 | 0.00 | 0.00 | 1,099.20 | |

## Field 6 - 10/02/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough Plant date: 10/02/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/22/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 106.69 | 14.21 | 575.95 | 2,492.63 | 4,356,000.00 *gal* |
| Application event totals | | 106.69 | 14.21 | 575.95 | 2,492.63 | |

| | | | | |
|---|---|---|---|---|
| 01/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 70.04 | 7.49 | 441.97 | 1,835.29 | 3,456,000.00 *gal* |
| Irrigation Wells | Ground water | 3.15 | 0.00 | 0.00 | 252.70 | 5,299,200.00 *gal* |
| Irrigation Wells | Ground water | 14.14 | 0.00 | 0.00 | 1,058.10 | 8,640,000.00 *gal* |
| Application event totals | | 87.33 | 7.49 | 441.97 | 3,146.09 | |

## Field 6 - 05/15/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage Plant date: 05/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

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## Field 6 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/16/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 68.98 | 9.77 | 29.30 | 0.00 | 918.00 *ton* |
| Application event totals | | 68.98 | 9.77 | 29.30 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.92 | 0.00 | 0.00 | 234.27 | 4,912,800.00 *gal* |
| Irrigation Wells | Ground water | 13.11 | 0.00 | 0.00 | 980.95 | 8,010,000.00 *gal* |
| Application event totals | | 16.03 | 0.00 | 0.00 | 1,215.22 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 141.33 | 10.83 | 185.52 | 994.90 | 2,700,000.00 *gal* |
| Irrigation Wells | Ground water | 2.46 | 0.00 | 0.00 | 197.42 | 4,140,000.00 *gal* |
| Irrigation Wells | Ground water | 11.05 | 0.00 | 0.00 | 826.64 | 6,750,000.00 *gal* |
| Application event totals | | 154.84 | 10.83 | 185.52 | 2,018.96 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.40 | 0.00 | 0.00 | 192.15 | 4,029,600.00 *gal* |
| Irrigation Wells | Ground water | 10.75 | 0.00 | 0.00 | 804.60 | 6,570,000.00 *gal* |
| Application event totals | | 13.15 | 0.00 | 0.00 | 996.75 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/07/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 41.55 | 5.43 | 293.63 | 1,246.55 | 2,556,000.00 *gal* |
| Irrigation Wells | Ground water | 2.33 | 0.00 | 0.00 | 186.89 | 3,919,200.00 *gal* |
| Irrigation Wells | Ground water | 10.46 | 0.00 | 0.00 | 782.56 | 6,390,000.00 *gal* |
| Application event totals | | 54.34 | 5.43 | 293.63 | 2,215.99 | |

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## Field 6 - 05/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.53 | 0.00 | 0.00 | 202.68 | 4,250,400.00 *gal* |
| Irrigation Wells | Ground water | 11.34 | 0.00 | 0.00 | 848.69 | 6,930,000.00 *gal* |
| Application event totals | | 13.87 | 0.00 | 0.00 | 1,051.37 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.17 | 0.00 | 0.00 | 173.73 | 3,643,200.00 *gal* |
| Irrigation Wells | Ground water | 9.72 | 0.00 | 0.00 | 727.45 | 5,940,000.00 *gal* |
| Application event totals | | 11.89 | 0.00 | 0.00 | 901.17 | |

## Field 7 - 10/10/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/15/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Compost | 237.43 | 22.68 | 60.02 | 0.00 | 540.00 *ton* |
| Application event totals | | 237.43 | 22.68 | 60.02 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/22/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.27 | 0.00 | 0.00 | 507.12 | 5,976,000.00 *gal* |
| Application event totals | | 1.27 | 0.00 | 0.00 | 507.12 | |

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## Field 7 - 10/10/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 02/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.02 | 0.00 | 0.00 | 409.36 | 4,824,000.00 *gal* |
| Application event totals | | 1.02 | 0.00 | 0.00 | 409.36 | |

## Field 7 - 05/22/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 05/22/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/26/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Solid Manure | Separator solids | 153.73 | 21.77 | 65.30 | 0.00 | 2,046.00 *ton* |
| Application event totals | | 153.73 | 21.77 | 65.30 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 05/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.16 | 0.00 | 0.00 | 861.49 | 10,152,000.00 *gal* |
| Application event totals | | 2.16 | 0.00 | 0.00 | 861.49 | |

| | | | | |
|---|---|---|---|---|
| 06/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.06 | 0.00 | 0.00 | 824.83 | 9,720,000.00 *gal* |
| Application event totals | | 2.06 | 0.00 | 0.00 | 824.83 | |

| | | | | |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.88 | 0.00 | 0.00 | 751.51 | 8,856,000.00 *gal* |
| Application event totals | | 1.88 | 0.00 | 0.00 | 751.51 | |

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## Field 7 - 05/22/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/08/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.11 | 0.00 | 0.00 | 843.16 | 9,936,000.00 *gal* |
| Application event totals | | 2.11 | 0.00 | 0.00 | 843.16 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.84 | 0.00 | 0.00 | 733.18 | 8,640,000.00 *gal* |
| Application event totals | | 1.84 | 0.00 | 0.00 | 733.18 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.00 | 0.00 | 0.00 | 800.39 | 9,432,000.00 *gal* |
| Application event totals | | 2.00 | 0.00 | 0.00 | 800.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.62 | 0.00 | 0.00 | 647.64 | 7,632,000.00 *gal* |
| Application event totals | | 1.62 | 0.00 | 0.00 | 647.64 | |

## Field 8 - 10/01/2020: Alfalfa, hay

Field name: Field 8

Crop: Alfalfa, hay

Plant date: 10/01/2020

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.58 | 0.00 | 0.00 | 1,032.39 | 11,376,000.00 *gal* |
| Application event totals | | 2.58 | 0.00 | 0.00 | 1,032.39 | |

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Field 8 - 10/01/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 10-10-10-9S | Solid commercial fertilizer | 100.00 | 100.00 | 100.00 | 0.00 | |
| Irrigation Wells | Ground water | 2.16 | 0.00 | 0.00 | 862.51 | 9,504,000.00 *gal* |
| Application event totals | | 102.16 | 100.00 | 100.00 | 862.51 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.50 | 0.00 | 0.00 | 999.72 | 11,016,000.00 *gal* |
| Application event totals | | 2.50 | 0.00 | 0.00 | 999.72 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.26 | 0.00 | 0.00 | 901.71 | 9,936,000.00 *gal* |
| Application event totals | | 2.26 | 0.00 | 0.00 | 901.71 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.22 | 0.00 | 0.00 | 888.64 | 9,792,000.00 *gal* |
| Application event totals | | 2.22 | 0.00 | 0.00 | 888.64 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.00 | 0.00 | 0.00 | 797.16 | 8,784,000.00 *gal* |
| Application event totals | | 2.00 | 0.00 | 0.00 | 797.16 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.04 | 0.00 | 0.00 | 816.77 | 9,000,000.00 *gal* |
| Application event totals | | 2.04 | 0.00 | 0.00 | 816.77 | |

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## Field 8 - 10/01/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.09 | 0.00 | 0.00 | 836.37 | 9,216,000.00 *gal* |
| Application event totals | | 2.09 | 0.00 | 0.00 | 836.37 | |

## Field 9 - 01/01/2020: Alfalfa, hay

Field name: Field 9

Crop: Alfalfa, hay

Plant date: 01/01/2020

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 137.57 | 10.54 | 180.57 | 968.37 | 2,628,000.00 *gal* |
| Irrigation Wells | Ground water | 2.40 | 0.00 | 0.00 | 192.15 | 4,029,600.00 *gal* |
| Irrigation Wells | Ground water | 10.75 | 0.00 | 0.00 | 804.60 | 6,570,000.00 *gal* |
| Application event totals | | 150.71 | 10.54 | 180.57 | 1,965.12 | |

| | | | | |
|---|---|---|---|---|
| 05/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.97 | 0.00 | 0.00 | 157.94 | 3,312,000.00 *gal* |
| Irrigation Wells | Ground water | 8.84 | 0.00 | 0.00 | 661.31 | 5,400,000.00 *gal* |
| Application event totals | | 10.81 | 0.00 | 0.00 | 819.25 | |

| | | | | |
|---|---|---|---|---|
| 06/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| 10-10-10-9S | Solid commercial fertilizer | 50.00 | 50.00 | 50.00 | 0.00 | |
| Irrigation Wells | Ground water | 2.17 | 0.00 | 0.00 | 173.73 | 3,643,200.00 *gal* |
| Irrigation Wells | Ground water | 9.72 | 0.00 | 0.00 | 727.45 | 5,940,000.00 *gal* |
| Application event totals | | 61.89 | 50.00 | 50.00 | 901.17 | |

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Field 9 - 01/01/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 2.00 | 0.00 | 0.00 | 160.57 | 3,367,200.00 *gal* |
| Irrigation Wells | Ground water | 8.98 | 0.00 | 0.00 | 672.34 | 5,490,000.00 *gal* |
| Application event totals | | 10.99 | 0.00 | 0.00 | 832.90 | |

| | | | | |
|---|---|---|---|---|
| 07/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 4.83 | 0.00 | 0.00 | 386.94 | 8,114,400.00 *gal* |
| Application event totals | | 4.83 | 0.00 | 0.00 | 386.94 | |

| | | | | |
|---|---|---|---|---|
| 08/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.94 | 0.00 | 0.00 | 155.30 | 3,256,800.00 *gal* |
| Irrigation Wells | Ground water | 8.69 | 0.00 | 0.00 | 650.29 | 5,310,000.00 *gal* |
| Application event totals | | 10.63 | 0.00 | 0.00 | 805.60 | |

| | | | | |
|---|---|---|---|---|
| 08/21/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.84 | 0.00 | 0.00 | 147.41 | 3,091,200.00 *gal* |
| Irrigation Wells | Ground water | 8.25 | 0.00 | 0.00 | 617.23 | 5,040,000.00 *gal* |
| Application event totals | | 10.09 | 0.00 | 0.00 | 764.63 | |

| | | | | |
|---|---|---|---|---|
| 09/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.90 | 0.00 | 0.00 | 152.67 | 3,201,600.00 *gal* |
| Irrigation Wells | Ground water | 8.54 | 0.00 | 0.00 | 639.27 | 5,220,000.00 *gal* |
| Application event totals | | 10.45 | 0.00 | 0.00 | 791.94 | |

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Field 9 - 01/01/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| WWS | Process wastewater | 77.84 | 10.17 | 550.04 | 2,335.08 | 4,788,000.00 *gal* |
| Irrigation Wells | Ground water | 4.37 | 0.00 | 0.00 | 350.09 | 7,341,600.00 *gal* |
| Application event totals | | 82.20 | 10.17 | 550.04 | 2,685.17 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Wells | Ground water | 1.81 | 0.00 | 0.00 | 144.77 | 3,036,000.00 *gal* |
| Irrigation Wells | Ground water | 8.10 | 0.00 | 0.00 | 606.20 | 4,950,000.00 *gal* |
| Application event totals | | 9.91 | 0.00 | 0.00 | 750.98 | |

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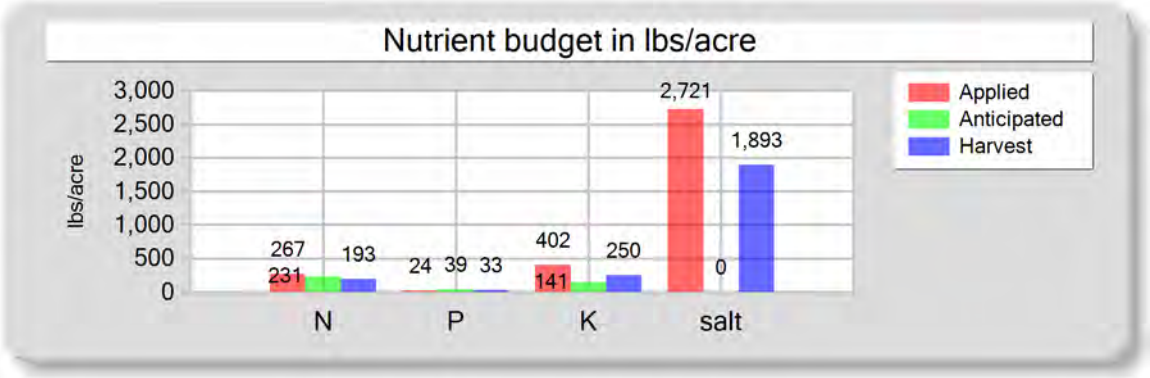
## B. NUTRIENT BUDGET

Field 1 - 10/01/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/01/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 187.33 | 17.89 | 47.36 | 0.00 |
| Process wastewater | 56.18 | 6.01 | 354.50 | 1,472.06 |
| Fresh water | 16.66 | 0.00 | 0.00 | 1,249.39 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 267.17 | 23.90 | 401.86 | 2,721.44 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 193.33 | 33.33 | 250.00 | 1,892.53 |
| Nutrient balance | 73.84 | -9.43 | 151.86 | 828.91 |
| Applied to removed ratio | 1.38 | 0.72 | 1.61 | 1.44 |

| Fresh water applied | |
|---|---|
| 2,023,200.00 | *gallons* |
| 74.51 | *acre-inches* |
| 4.97 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 540,000.00 | *gallons* |
| 19.89 | *acre-inches* |
| 1.33 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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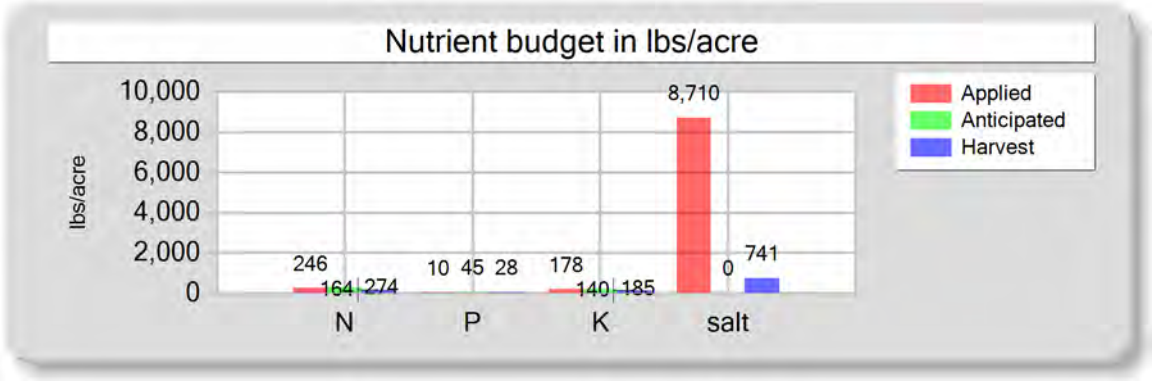
## Field 1 - 05/17/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/17/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 135.43 | 10.37 | 177.77 | 953.33 |
| Fresh water | 103.42 | 0.00 | 0.00 | 7,756.61 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 245.85 | 10.37 | 177.77 | 8,709.94 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 164.00 | 28.00 | 140.00 | 740.88 |
| Nutrient balance | 81.85 | -17.63 | 37.77 | 7,969.06 |
| Applied to removed ratio | 1.50 | 0.37 | 1.27 | 11.76 |

| Fresh water applied | |
|---|---|
| 12,560,700.00 | *gallons* |
| 462.57 | *acre-inches* |
| 30.84 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 504,000.00 | *gallons* |
| 18.56 | *acre-inches* |
| 1.24 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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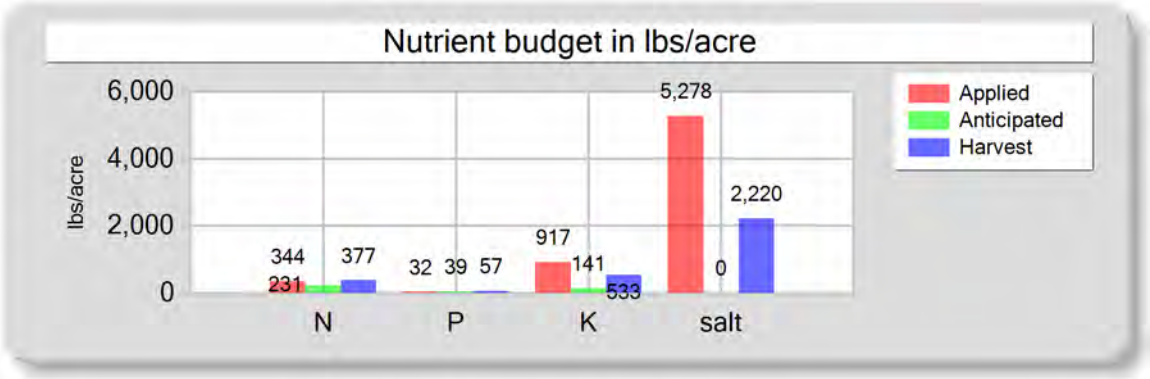
## Field 11 - 10/20/2022: Rye Silage

Field name: Field 11 Crop: Rye Silage Plant date: 10/20/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 176.66 | 16.87 | 44.66 | 0.00 |
| Process wastewater | 138.29 | 14.79 | 872.60 | 3,623.53 |
| Fresh water | 22.26 | 0.00 | 0.00 | 1,654.55 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 344.20 | 31.66 | 917.26 | 5,278.07 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 376.67 | 56.67 | 533.33 | 2,220.00 |
| Nutrient balance | -32.47 | -25.01 | 383.93 | 3,058.07 |
| Applied to removed ratio | 0.91 | 0.56 | 1.72 | 2.38 |

| Fresh water applied | |
|---|---|
| 13,685,760.00 | *gallons* |
| 504.00 | *acre-inches* |
| 6.46 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,912,000.00 | *gallons* |
| 254.55 | *acre-inches* |
| 3.26 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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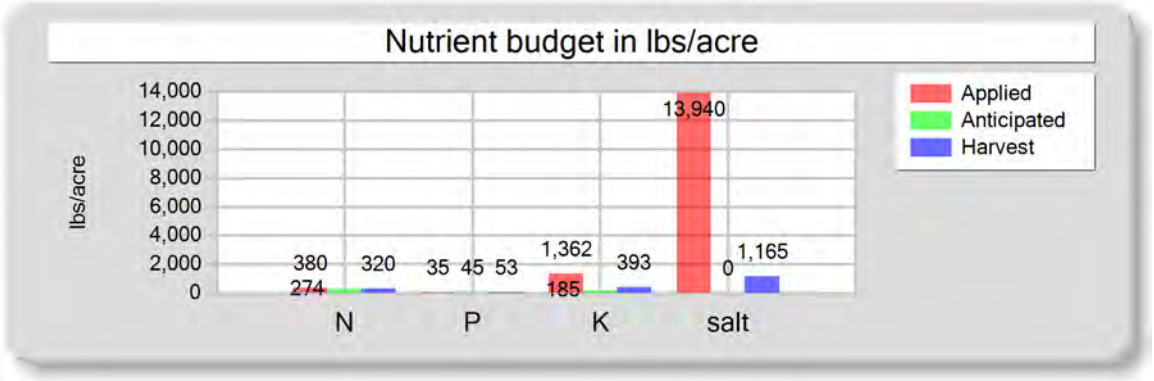
## Field 11 - 06/16/2023: Corn, silage

Field name: Field 11 Crop: Corn, silage Plant date: 06/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 72.69 | 10.29 | 30.88 | 0.00 |
| Process wastewater | 188.34 | 24.61 | 1,330.94 | 5,650.21 |
| Fresh water | 111.51 | 0.00 | 0.00 | 8,289.97 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 379.54 | 34.90 | 1,361.82 | 13,940.18 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 320.00 | 53.33 | 393.33 | 1,164.80 |
| Nutrient balance | 59.54 | -18.43 | 968.48 | 12,775.38 |
| Applied to removed ratio | 1.19 | 0.65 | 3.46 | 11.97 |

| Fresh water applied | |
|---|---|
| 68,571,360.00 | *gallons* |
| 2,525.25 | *acre-inches* |
| 32.38 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 11,736,000.00 | *gallons* |
| 432.20 | *acre-inches* |
| 5.54 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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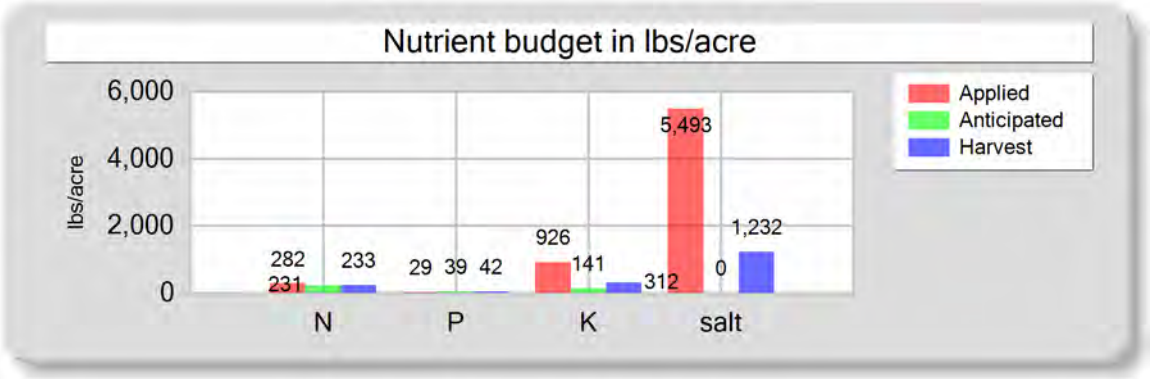
## Field 12 - 10/22/2022: Oats, silage-soft dough

Field name: Field 12 Crop: Oats, silage-soft dough Plant date: 10/22/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 97.04 | 10.21 | 26.17 | 0.00 |
| Process wastewater | 155.15 | 18.88 | 899.39 | 3,817.36 |
| Fresh water | 22.54 | 0.00 | 0.00 | 1,676.03 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 281.73 | 29.10 | 925.56 | 5,493.40 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 232.73 | 41.56 | 311.69 | 1,231.63 |
| Nutrient balance | 49.00 | -12.46 | 613.87 | 4,261.77 |
| Applied to removed ratio | 1.21 | 0.70 | 2.97 | 4.46 |

| Fresh water applied | |
|---|---|
| 13,685,760.00 | *gallons* |
| 504.00 | *acre-inches* |
| 6.55 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,912,000.00 | *gallons* |
| 254.55 | *acre-inches* |
| 3.31 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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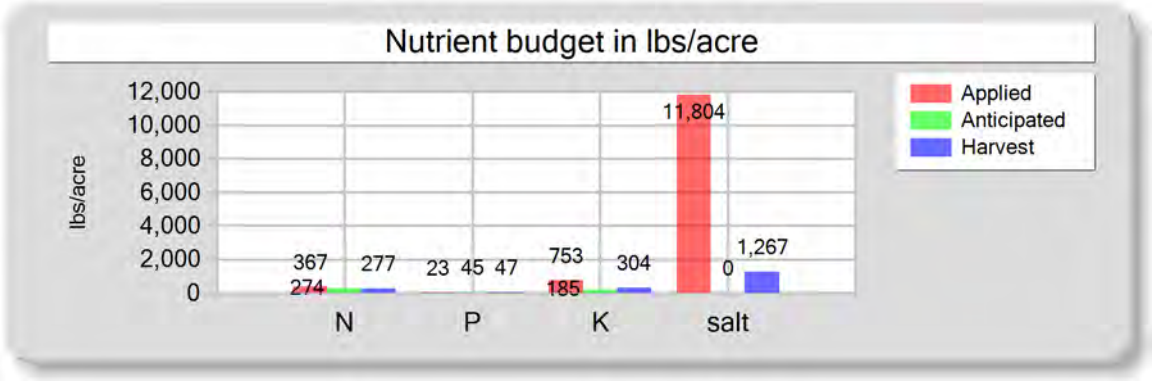
## Field 12 - 05/24/2023: Corn, silage

Field name: Field 12 Crop: Corn, silage Plant date: 05/24/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 247.69 | 22.99 | 752.80 | 3,450.15 |
| Fresh water | 112.37 | 0.00 | 0.00 | 8,353.98 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 367.07 | 22.99 | 752.80 | 11,804.13 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 276.88 | 47.27 | 303.90 | 1,266.91 |
| Nutrient balance | 90.18 | -24.28 | 448.90 | 10,537.22 |
| Applied to removed ratio | 1.33 | 0.49 | 2.48 | 9.32 |

| Fresh water applied | |
|---|---|
| 68,214,960.00 | *gallons* |
| 2,512.13 | *acre-inches* |
| 32.63 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 7,884,000.00 | *gallons* |
| 290.34 | *acre-inches* |
| 3.77 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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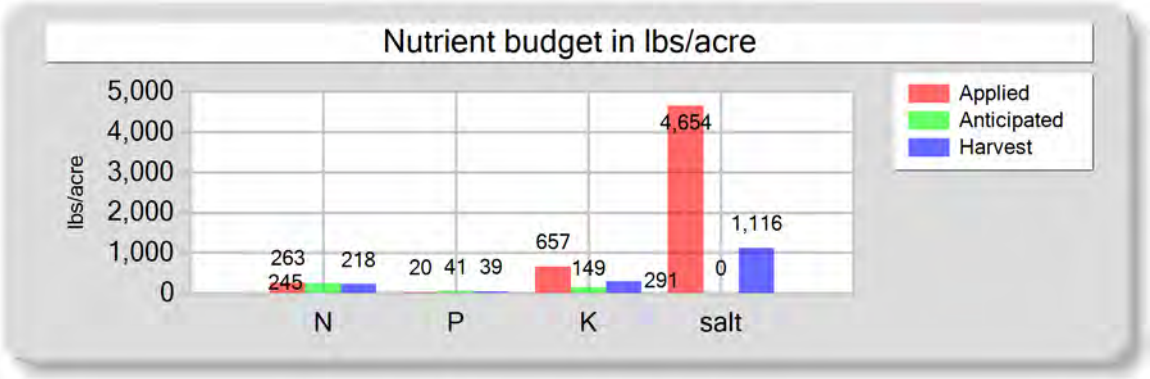
*Reporting period 01/01/2023 to 12/31/2023.*

## Field 14 - 11/01/2022: Oats, hay

Field name: Field 14 Crop: Oats, hay Plant date: 11/01/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 233.46 | 20.02 | 657.39 | 2,989.18 |
| Fresh water | 22.30 | 0.00 | 0.00 | 1,664.83 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 262.76 | 20.02 | 657.39 | 4,654.01 |
| Anticipated crop nutrient removal | 244.80 | 41.40 | 149.40 | 0.00 |
| Actual crop nutrient removal | 218.00 | 39.00 | 291.00 | 1,116.00 |
| Nutrient balance | 44.76 | -18.98 | 366.39 | 3,538.01 |
| Applied to removed ratio | 1.21 | 0.51 | 2.26 | 4.17 |

| Fresh water applied | |
|---|---|
| 14,250,000.00 | *gallons* |
| 524.78 | *acre-inches* |
| 6.56 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,840,000.00 | *gallons* |
| 251.89 | *acre-inches* |
| 3.15 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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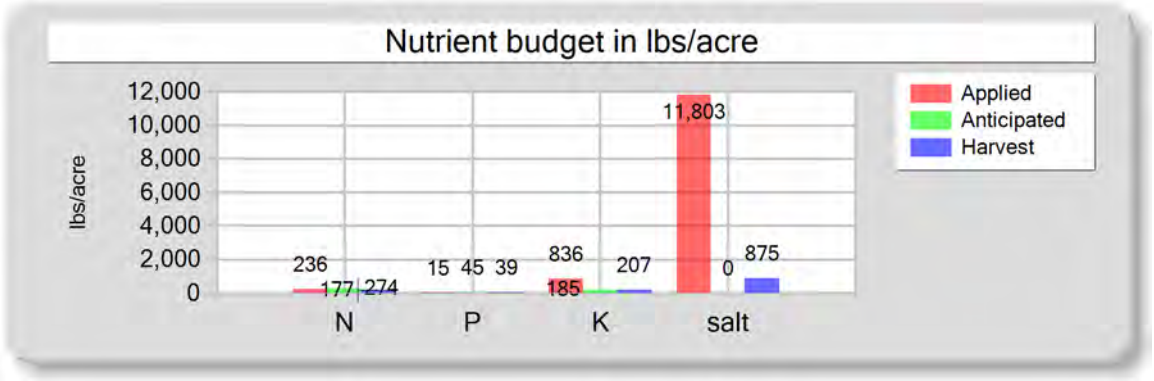
## Field 14 - 07/04/2023: Corn, silage

Field name: Field 14 Crop: Corn, silage Plant date: 07/04/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 118.29 | 15.46 | 835.92 | 3,548.71 |
| Fresh water | 110.55 | 0.00 | 0.00 | 8,254.04 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 235.84 | 15.46 | 835.92 | 11,802.75 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 177.12 | 39.36 | 206.64 | 875.37 |
| Nutrient balance | 58.72 | -23.90 | 629.28 | 10,927.38 |
| Applied to removed ratio | 1.33 | 0.39 | 4.05 | 13.48 |

| Fresh water applied | |
|---|---|
| 70,650,000.00 | *gallons* |
| 2,601.80 | *acre-inches* |
| 32.52 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 7,560,000.00 | *gallons* |
| 278.41 | *acre-inches* |
| 3.48 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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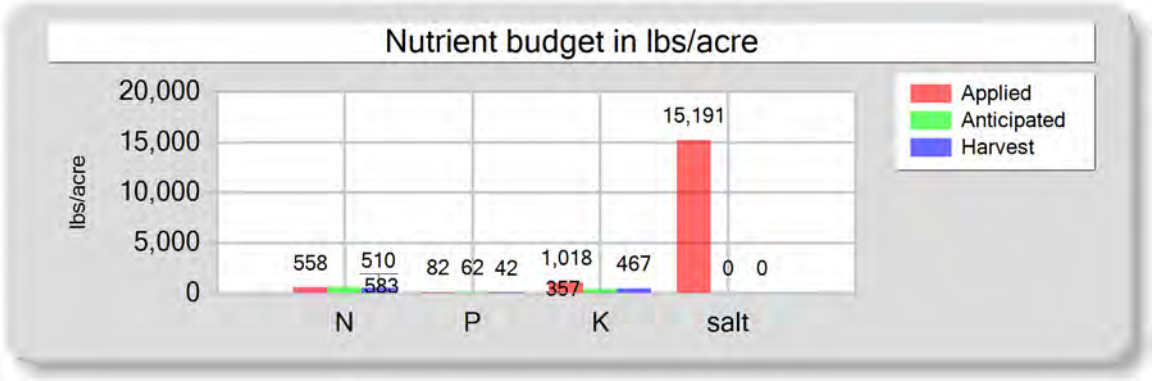
## Field 15 - 09/20/2022: Alfalfa, hay

Field name: Field 15 Crop: Alfalfa, hay Plant date: 09/20/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 50.00 | 50.00 | 50.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 350.24 | 31.60 | 967.52 | 4,491.70 |
| Fresh water | 143.30 | 0.00 | 0.00 | 10,699.68 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 557.54 | 81.60 | 1,017.52 | 15,191.38 |
| Anticipated crop nutrient removal | 583.00 | 62.00 | 357.00 | 0.00 |
| Actual crop nutrient removal | 510.22 | 42.00 | 466.67 | 0.00 |
| Nutrient balance | 47.32 | 39.60 | 550.85 | 15,191.38 |
| Applied to removed ratio | 1.09 | 1.94 | 2.18 | 0.00 |

| Fresh water applied | |
|---|---|
| 82,425,000.00 | *gallons* |
| 3,035.43 | *acre-inches* |
| 42.16 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 9,756,000.00 | *gallons* |
| 359.28 | *acre-inches* |
| 4.99 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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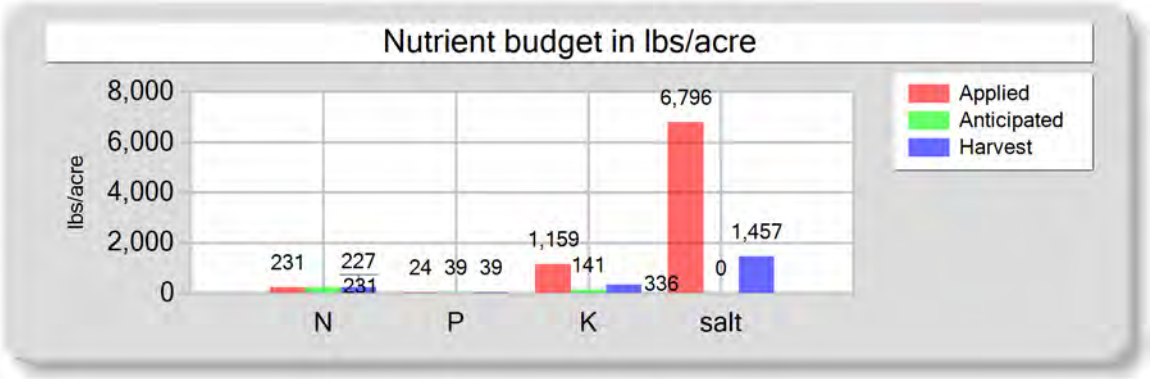
## Field 2 - 10/03/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/03/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 198.05 | 23.80 | 1,158.53 | 4,905.48 |
| Fresh water | 25.48 | 0.00 | 0.00 | 1,890.11 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 230.53 | 23.80 | 1,158.53 | 6,795.58 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 226.91 | 39.27 | 336.00 | 1,456.76 |
| Nutrient balance | 3.62 | -15.47 | 822.53 | 5,338.83 |
| Applied to removed ratio | 1.02 | 0.61 | 3.45 | 4.66 |

| Fresh water applied | |
|---|---|
| 11,122,560.00 | *gallons* |
| 409.61 | *acre-inches* |
| 7.45 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,372,000.00 | *gallons* |
| 234.66 | *acre-inches* |
| 4.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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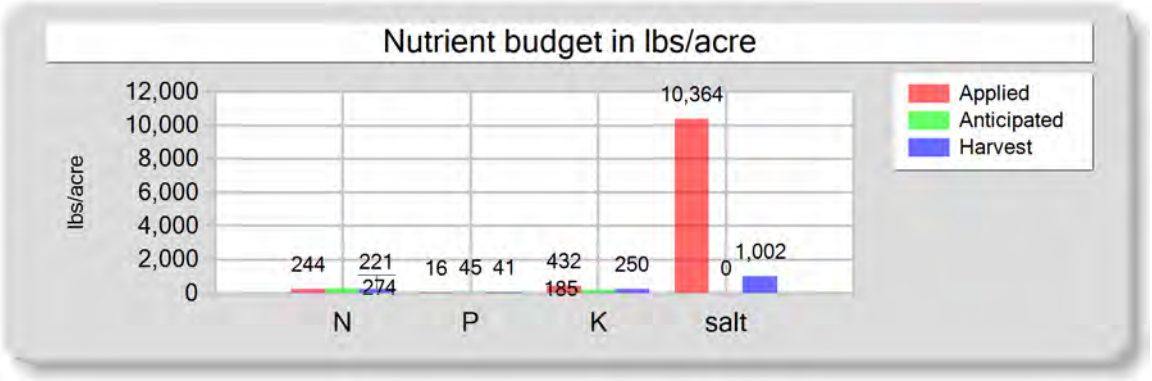
## Field 2 - 05/25/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/25/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 62.69 | 8.88 | 26.63 | 0.00 |
| Process wastewater | 57.35 | 7.49 | 405.29 | 1,720.59 |
| Fresh water | 116.53 | 0.00 | 0.00 | 8,643.30 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 243.58 | 16.37 | 431.93 | 10,363.89 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 221.09 | 40.73 | 250.18 | 1,001.54 |
| Nutrient balance | 22.49 | -24.36 | 181.74 | 9,362.35 |
| Applied to removed ratio | 1.10 | 0.40 | 1.73 | 10.35 |

| Fresh water applied | |
|---|---|
| 50,862,540.00 | *gallons* |
| 1,873.09 | *acre-inches* |
| 34.06 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,520,000.00 | *gallons* |
| 92.80 | *acre-inches* |
| 1.69 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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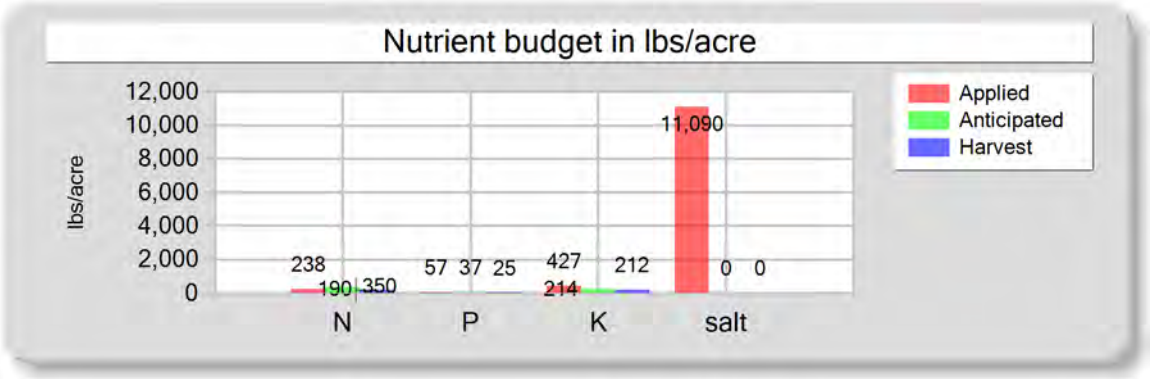
## Field 3 - 09/10/2015: Alfalfa, hay

Field name: Field 3 Crop: Alfalfa, hay Plant date: 09/10/2015




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 50.00 | 50.00 | 50.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 53.36 | 6.97 | 377.11 | 1,600.92 |
| Fresh water | 127.94 | 0.00 | 0.00 | 9,489.40 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 238.30 | 56.97 | 427.11 | 11,090.32 |
| Anticipated crop nutrient removal | 349.80 | 37.20 | 214.20 | 0.00 |
| Actual crop nutrient removal | 189.74 | 24.87 | 211.84 | 0.00 |
| Nutrient balance | 48.56 | 32.10 | 215.26 | 11,090.32 |
| Applied to removed ratio | 1.26 | 2.29 | 2.02 | 0.00 |

| Fresh water applied | |
|---|---|
| 77,162,760.00 | *gallons* |
| 2,841.64 | *acre-inches* |
| 37.39 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,240,000.00 | *gallons* |
| 119.32 | *acre-inches* |
| 1.57 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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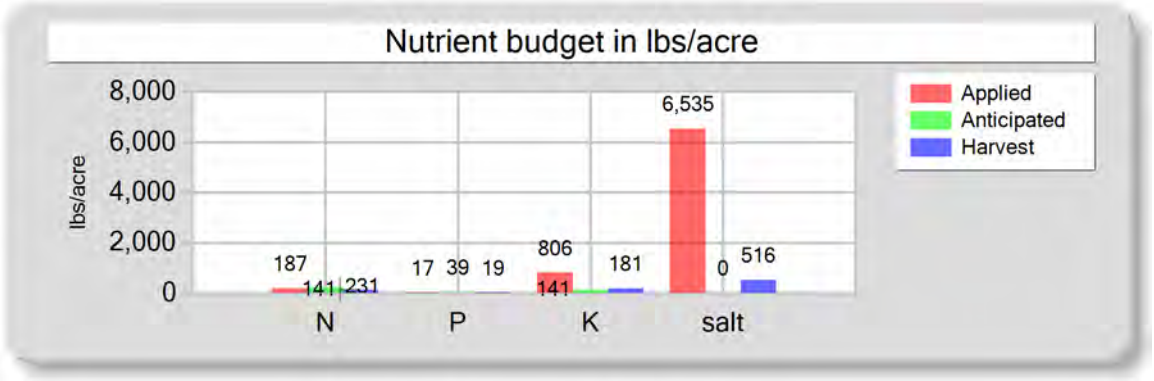
## Field 3 - 09/30/2022: Rye Silage

Field name: Field 3 Crop: Rye Silage Plant date: 09/30/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 138.15 | 16.67 | 805.76 | 3,414.43 |
| Fresh water | 42.07 | 0.00 | 0.00 | 3,120.39 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 187.22 | 16.67 | 805.76 | 6,534.82 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 141.47 | 18.95 | 180.63 | 515.77 |
| Nutrient balance | 45.75 | -2.27 | 625.13 | 6,019.05 |
| Applied to removed ratio | 1.32 | 0.88 | 4.46 | 12.67 |

| Fresh water applied | |
|---|---|
| 25,373,340.00 | *gallons* |
| 934.41 | *acre-inches* |
| 12.29 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,120,000.00 | *gallons* |
| 225.38 | *acre-inches* |
| 2.97 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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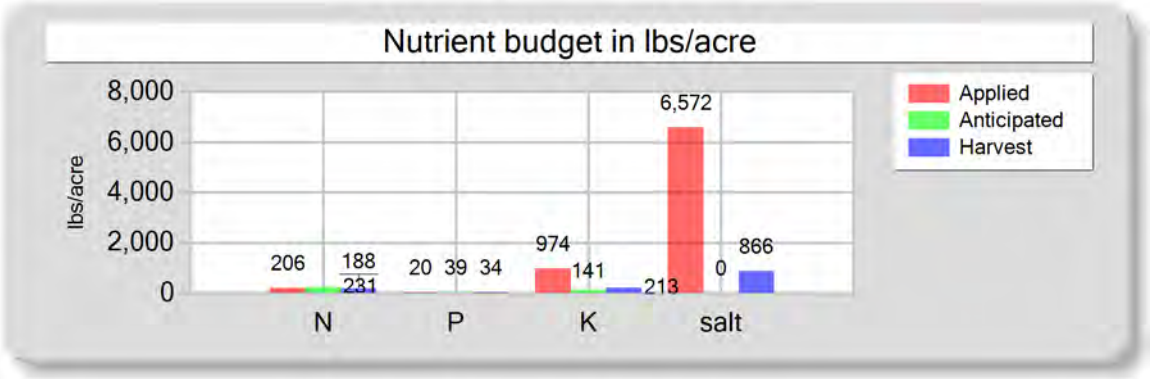
## Field 4 - 10/02/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/02/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 167.10 | 20.18 | 974.21 | 4,128.70 |
| Fresh water | 32.23 | 0.00 | 0.00 | 2,443.71 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 206.33 | 20.18 | 974.21 | 6,572.41 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 188.03 | 33.80 | 213.38 | 865.90 |
| Nutrient balance | 18.30 | -13.62 | 760.83 | 5,706.51 |
| Applied to removed ratio | 1.10 | 0.60 | 4.57 | 7.59 |

| Fresh water applied | |
|---|---|
| 24,998,400.00 | *gallons* |
| 920.61 | *acre-inches* |
| 12.97 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,912,000.00 | *gallons* |
| 254.55 | *acre-inches* |
| 3.59 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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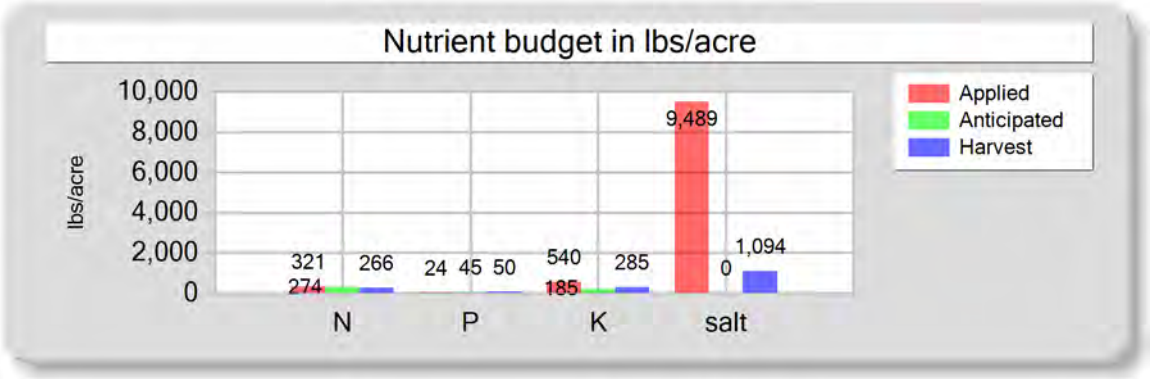
## Field 4 - 05/16/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 65.19 | 9.23 | 27.69 | 0.00 |
| Process wastewater | 154.09 | 14.72 | 512.69 | 2,323.39 |
| Fresh water | 94.50 | 0.00 | 0.00 | 7,165.67 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 320.78 | 23.95 | 540.38 | 9,489.06 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 266.48 | 49.58 | 285.07 | 1,094.11 |
| Nutrient balance | 54.30 | -25.63 | 255.31 | 8,394.94 |
| Applied to removed ratio | 1.20 | 0.48 | 1.90 | 8.67 |

| Fresh water applied | |
|---|---|
| 73,302,600.00 | *gallons* |
| 2,699.49 | *acre-inches* |
| 38.02 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 4,824,000.00 | *gallons* |
| 177.65 | *acre-inches* |
| 2.50 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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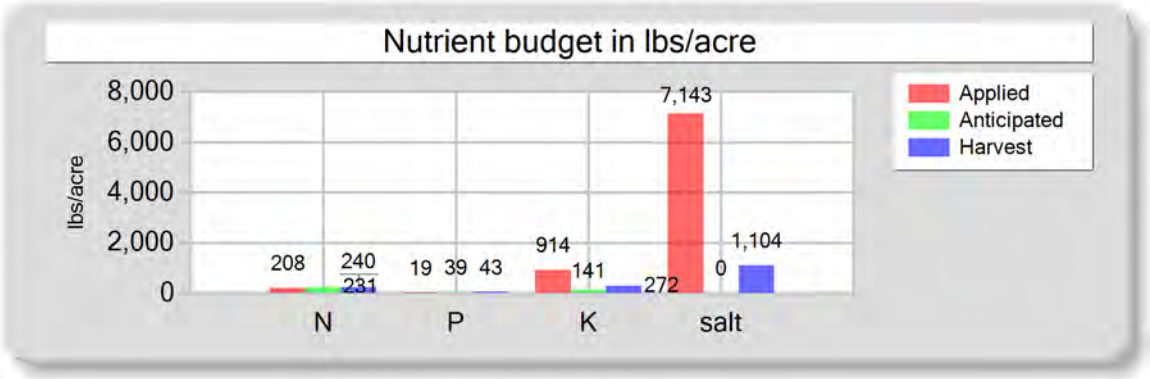
## Field 5 - 10/10/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 10/10/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 157.78 | 19.23 | 913.92 | 3,879.90 |
| Fresh water | 43.57 | 0.00 | 0.00 | 3,263.24 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 208.36 | 19.23 | 913.92 | 7,143.14 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 239.71 | 43.09 | 272.03 | 1,103.89 |
| Nutrient balance | -31.35 | -23.87 | 641.89 | 6,039.25 |
| Applied to removed ratio | 0.87 | 0.45 | 3.36 | 6.47 |

| Fresh water applied | |
|---|---|
| 26,185,800.00 | *gallons* |
| 964.33 | *acre-inches* |
| 12.86 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,840,000.00 | *gallons* |
| 251.89 | *acre-inches* |
| 3.36 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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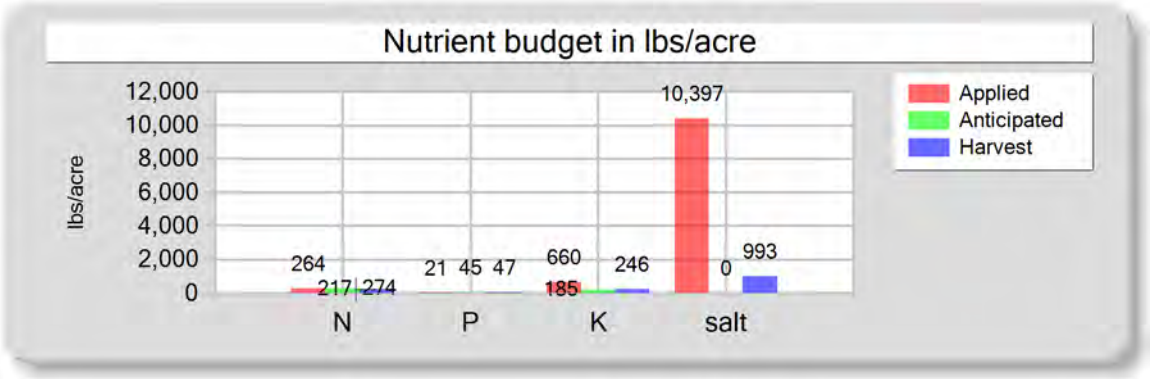
## Field 5 - 06/26/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/26/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 64.64 | 9.15 | 27.46 | 0.00 |
| Process wastewater | 89.53 | 11.70 | 632.64 | 2,685.75 |
| Fresh water | 102.97 | 0.00 | 0.00 | 7,711.55 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 264.14 | 20.85 | 660.10 | 10,397.30 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 217.07 | 46.93 | 246.40 | 993.40 |
| Nutrient balance | 47.07 | -26.08 | 413.70 | 9,403.90 |
| Applied to removed ratio | 1.22 | 0.44 | 2.68 | 10.47 |

| Fresh water applied | |
|---|---|
| 61,881,180.00 | *gallons* |
| 2,278.87 | *acre-inches* |
| 30.38 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 5,364,000.00 | *gallons* |
| 197.54 | *acre-inches* |
| 2.63 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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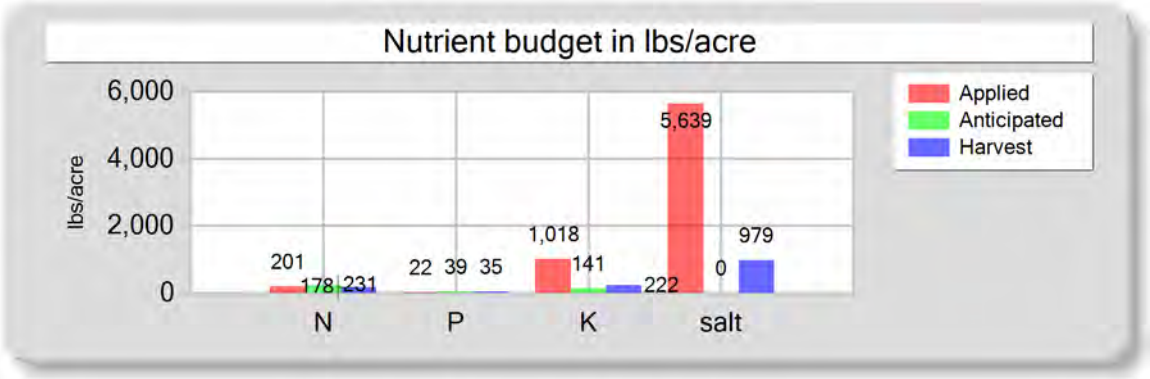
## Field 6 - 10/02/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 10/02/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 176.73 | 21.70 | 1,017.92 | 4,327.92 |
| Fresh water | 17.29 | 0.00 | 0.00 | 1,310.80 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 201.03 | 21.70 | 1,017.92 | 5,638.72 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 177.66 | 35.06 | 222.08 | 979.36 |
| Nutrient balance | 23.36 | -13.36 | 795.84 | 4,659.36 |
| Applied to removed ratio | 1.13 | 0.62 | 4.58 | 5.76 |

| Fresh water applied | |
|---|---|
| 13,939,200.00 | *gallons* |
| 513.33 | *acre-inches* |
| 6.67 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 7,812,000.00 | *gallons* |
| 287.69 | *acre-inches* |
| 3.74 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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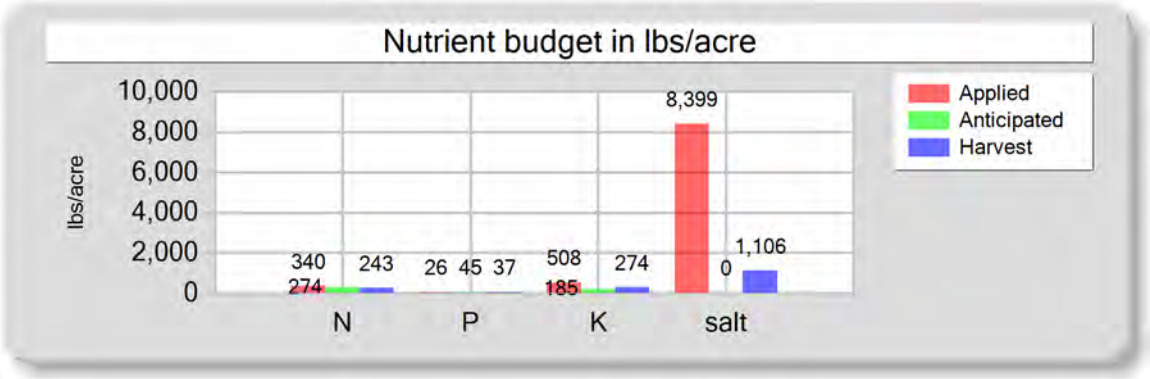
## Field 6 - 05/15/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 68.98 | 9.77 | 29.30 | 0.00 |
| Process wastewater | 182.89 | 16.26 | 479.15 | 2,241.45 |
| Fresh water | 81.24 | 0.00 | 0.00 | 6,158.02 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 340.10 | 26.02 | 508.45 | 8,399.47 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 243.12 | 37.40 | 274.29 | 1,106.37 |
| Nutrient balance | 96.98 | -11.38 | 234.16 | 7,293.10 |
| Applied to removed ratio | 1.40 | 0.70 | 1.85 | 7.59 |

| Fresh water applied | |
|---|---|
| 65,485,200.00 | *gallons* |
| 2,411.60 | *acre-inches* |
| 31.32 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 5,256,000.00 | *gallons* |
| 193.56 | *acre-inches* |
| 2.51 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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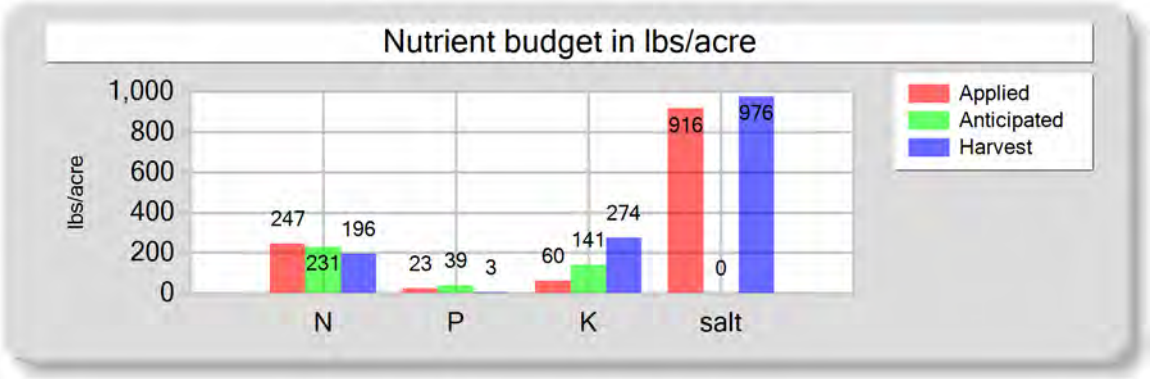
## Field 7 - 10/10/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 10/10/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 237.43 | 22.68 | 60.02 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 2.29 | 0.00 | 0.00 | 916.48 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 246.72 | 22.68 | 60.02 | 916.48 |
| Anticipated crop nutrient removal | 231.20 | 39.10 | 141.10 | 0.00 |
| Actual crop nutrient removal | 195.84 | 3.38 | 273.51 | 976.49 |
| Nutrient balance | 50.88 | 19.30 | -213.48 | -60.01 |
| Applied to removed ratio | 1.26 | 6.72 | 0.22 | 0.94 |

| Fresh water applied | |
|---|---|
| 10,800,000.00 | *gallons* |
| 397.73 | *acre-inches* |
| 5.17 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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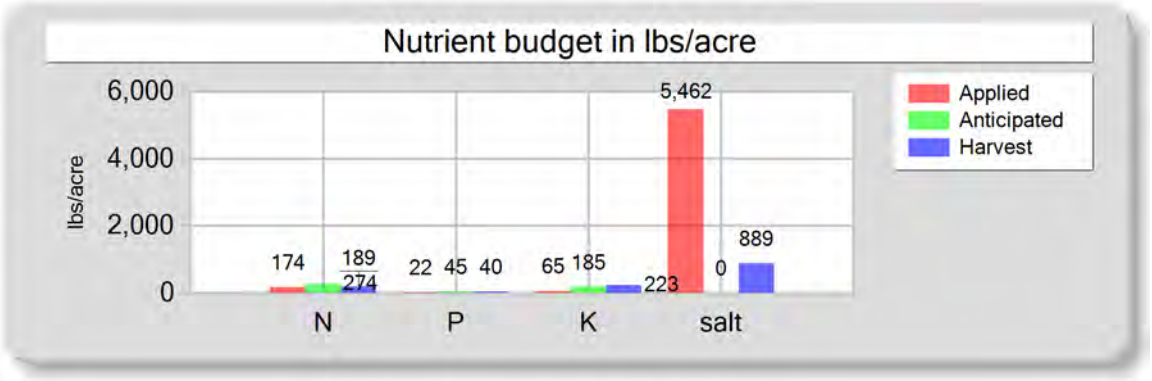
## Field 7 - 05/22/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 05/22/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 153.73 | 21.77 | 65.30 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 13.67 | 0.00 | 0.00 | 5,462.20 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 174.40 | 21.77 | 65.30 | 5,462.20 |
| Anticipated crop nutrient removal | 274.40 | 44.80 | 184.80 | 0.00 |
| Actual crop nutrient removal | 188.57 | 40.00 | 222.86 | 888.69 |
| Nutrient balance | -14.17 | -18.23 | -157.56 | 4,573.51 |
| Applied to removed ratio | 0.92 | 0.54 | 0.29 | 6.15 |

| Fresh water applied | |
|---|---|
| 64,368,000.00 | *gallons* |
| 2,370.45 | *acre-inches* |
| 30.79 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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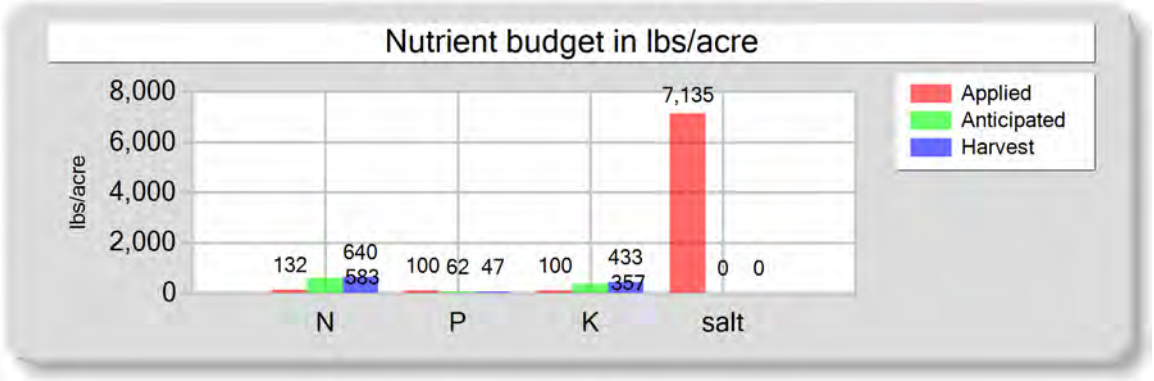
## Field 8 - 10/01/2020: Alfalfa, hay

Field name: Field 8 Crop: Alfalfa, hay Plant date: 10/01/2020




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 100.00 | 100.00 | 100.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 17.86 | 0.00 | 0.00 | 7,135.28 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 131.86 | 100.00 | 100.00 | 7,135.28 |
| Anticipated crop nutrient removal | 583.00 | 62.00 | 357.00 | 0.00 |
| Actual crop nutrient removal | 640.00 | 46.67 | 433.33 | 0.00 |
| Nutrient balance | -508.14 | 53.33 | -333.33 | 7,135.28 |
| Applied to removed ratio | 0.21 | 2.14 | 0.23 | 0.00 |

| Fresh water applied | |
|---|---|
| 78,624,000.00 | *gallons* |
| 2,895.45 | *acre-inches* |
| 40.21 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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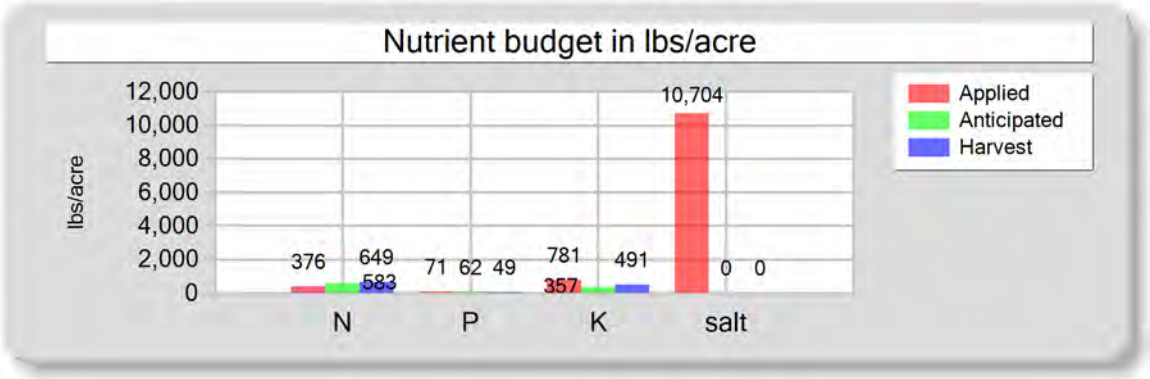
## Field 9 - 01/01/2020: Alfalfa, hay

Field name: Field 9 Crop: Alfalfa, hay Plant date: 01/01/2020




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 50.00 | 50.00 | 50.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 215.40 | 20.71 | 730.61 | 3,303.45 |
| Fresh water | 97.10 | 0.00 | 0.00 | 7,400.26 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 376.50 | 70.71 | 780.61 | 10,703.71 |
| Anticipated crop nutrient removal | 583.00 | 62.00 | 357.00 | 0.00 |
| Actual crop nutrient removal | 649.09 | 49.09 | 490.91 | 0.00 |
| Nutrient balance | -272.59 | 21.62 | 289.70 | 10,703.71 |
| Applied to removed ratio | 0.58 | 1.44 | 1.59 | 0.00 |

| Fresh water applied | |
|---|---|
| 86,313,600.00 | *gallons* |
| 3,178.64 | *acre-inches* |
| 41.28 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 7,416,000.00 | *gallons* |
| 273.11 | *acre-inches* |
| 3.55 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

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NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Corral Solids

Sample and source description: Corral Solids
Sample date: 09/29/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 28.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 21,600.00 | 5,700.00 | 32,100.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

### Drying Solids

Sample and source description: Drying Solids
Sample date: 09/29/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 4.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 17,800.00 | 1,700.00 | 4,500.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

### Separator Solids

Sample and source description: Separator Solids
Sample date: 09/29/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 77.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15,200.00 | 1,600.00 | 4,100.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

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## Corral Solids

Sample and source description: Corral Solids

Sample date: 04/20/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 8,900.00 | 3,400.00 | 18,600.00 | 17,800.00 | 12,500.00 | 3,800.00 | 3,500.00 | 8,000.00 | | 70.10 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 1,000.00 | | 0.01 |

## Drying Solids

Sample and source description: Drying Solids

Sample date: 04/20/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 5,000.00 | 900.00 | 6,500.00 | 16,800.00 | 10,800.00 | 1,300.00 | 1,500.00 | 0.00 | | 71.00 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 1,000.00 | | 0.01 |

## Separator Solids

Sample and source description: Separator Solids

Sample date: 04/20/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 11,300.00 | 1,600.00 | 4,800.00 | | | | | | | 0.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | 0.01 |

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Corral Solids

Sample and source description: Corral Solids
Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 6.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 16,500.00 | 5,600.00 | 32,900.00 | 15,000.00 | 9,600.00 | 7,500.00 | 4,200.00 | 19,000.00 | | 51.20 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 1,000.00 | | 0.01 |

Separator Solids

Sample and source description: Separator Solids
Sample date: 10/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 73.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 16,300.00 | 4,000.00 | 22,800.00 | 13,700.00 | 7,200.00 | 5,200.00 | 3,500.00 | 11,000.00 | | 35.40 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | | 0.01 |

## B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW
Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.30

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 226.00 | 65.90 | 0.00 | 3.80 | 30.10 | 1,220.00 | | | | | | | | 8,410.00 | 5,280 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

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1QWW

Sample and source description: 1QWW

Sample date: 02/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.30

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 187.00 | 66.90 | 0.00 | 1.90 | 20.00 | 1,180.00 | | | | | | | | 8,020.00 | 4,900 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

2QWW

Sample and source description: 2QWW

Sample date: 06/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 483.00 | 229.00 | 0.00 | 1.10 | 37.00 | 634.00 | | | | | | | | 8,230.00 | 3,400 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

3QWW

Sample and source description: 3QWW

Sample date: 09/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 924.00 | 497.00 | 0.00 | 0.00 | 145.00 | 676.00 | 321.00 | 192.00 | 224.00 | 4,010.00 | 0.00 | 64.90 | 254.00 | 7,290.00 | 2,700 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | 0.10 | 0.10 | 1.00 | 10.00 | 1.00 | 0.50 | 0.20 | 10.00 | 10 |

3QWW

Sample and source description: 3QWW

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.50

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 150.00 | 117.00 | 0.00 | 0.60 | 19.60 | 1,060.00 | | | | | | | | 7,200.00 | 4,500 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

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4QWW

Sample and source description: 4QWW

Sample date: 10/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 359.00 | 264.00 | 0.00 | 1.30 | 22.70 | 435.00 | | | | | | | | 5,130.00 | 2,380 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## C. FRESH WATER ANALYSES

Irrigation Wells

IW #4

Sample description: IW #4

Sample date: 12/15/2022 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 1.96 | 0.00 | 1.80 | | | | | | | | 1,510.00 | 783 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

IW #1

Sample description: IW #1

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15.10 | 0.00 | 15.10 | | | | | | | | 1,830.00 | 1,120 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

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## Irrigation Wells

### IW #2

Sample description: IW #2

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 14.90 | 0.00 | 14.90 | | | | | | | | 1,830.00 | 1,120 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

### IW #3

Sample description: IW #3

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 5.49 | 0.00 | 5.49 | | | | | | | | 822.00 | 440 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

### IW #4

Sample description: IW #4

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 58.90 | 0.00 | 58.00 | | | | | | | | 3,910.00 | 2,280 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

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## Irrigation Wells

### IW #5

Sample description: IW #5

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15.10 | 0.00 | 15.00 | | | | | | | | 1,840.00 | 1,130 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

### IW #6

Sample description: IW #6

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15.20 | 0.00 | 14.80 | | | | | | | | 1,790.00 | 1,130 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

### IW #7

Sample description: IW #7

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 14.80 | 0.00 | 14.80 | | | | | | | | 1,820.00 | 1,110 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

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Irrigation Wells

IW #8

Sample description: IW #8

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 5.50 | 0.00 | 5.50 | | | | | | | | 817.00 | 490 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

IW #9

Sample description: IW #9

Sample date: 02/23/2024 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 15.00 | 0.00 | 15.00 | | | | | | | | 1,840.00 | 1,120 |
| **DL** | 1.00 | 0.50 | 0.10 | | | | | | | | 10.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

Field 1 - 10/01/2022: Oats, silage-soft dough

Oat Silage

Sample and source description: Oat Silage

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 5,800.00 | 1,000.00 | 7,500.00 | | 15.10 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 1 - 05/17/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,100.00 | 700.00 | 3,500.00 | | 5.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 11 - 10/20/2022: Rye Silage

### Rye Silage

Sample and source description: Rye Silage
Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,300.00 | 1,700.00 | 16,000.00 | | 18.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 11 - 06/16/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,800.00 | 800.00 | 5,900.00 | | 5.20 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 12 - 10/22/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 76.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 5,600.00 | 1,000.00 | 7,500.00 | | 12.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 12 - 05/24/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,100.00 | 700.00 | 4,500.00 | | 5.60 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 14 - 11/01/2022: Oats, hay

### Oat Hay

Sample and source description: Oat Hay
Sample date: 06/30/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 10.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 21,800.00 | 3,900.00 | 29,100.00 | | 12.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 14 - 07/04/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,600.00 | 800.00 | 4,200.00 | | 6.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 2 - 10/03/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 5,200.00 | 900.00 | 7,700.00 | | 10.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 2 - 05/25/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,800.00 | 700.00 | 4,300.00 | | 5.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 3 - 09/30/2022: Rye Silage

### Rye Silage

Sample and source description: Rye Silage
Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 29.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 22,400.00 | 3,000.00 | 28,600.00 | | 11.60 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 4 - 10/02/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 8,900.00 | 1,600.00 | 10,100.00 | | 13.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 4 - 05/16/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 08/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,300.00 | 800.00 | 4,600.00 | | 5.50 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 5 - 10/10/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 8,900.00 | 1,600.00 | 10,100.00 | | 13.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 5 - 06/26/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,700.00 | 800.00 | 4,200.00 | | 5.90 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 6 - 10/02/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 7,600.00 | 1,500.00 | 9,500.00 | | 13.30 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

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## Field 6 - 05/15/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,900.00 | 600.00 | 4,400.00 | | 5.80 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 7 - 10/10/2022: Oats, silage-soft dough

### Oat Silage

Sample and source description: Oat Silage
Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 76.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 5,800.00 | 100.00 | 8,100.00 | | 12.10 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 7 - 05/22/2023: Corn, silage

### Corn Silage

Sample and source description: Corn Silage
Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,300.00 | 700.00 | 3,900.00 | | 5.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

*No subsurface (tile) drainage analyses entered.*

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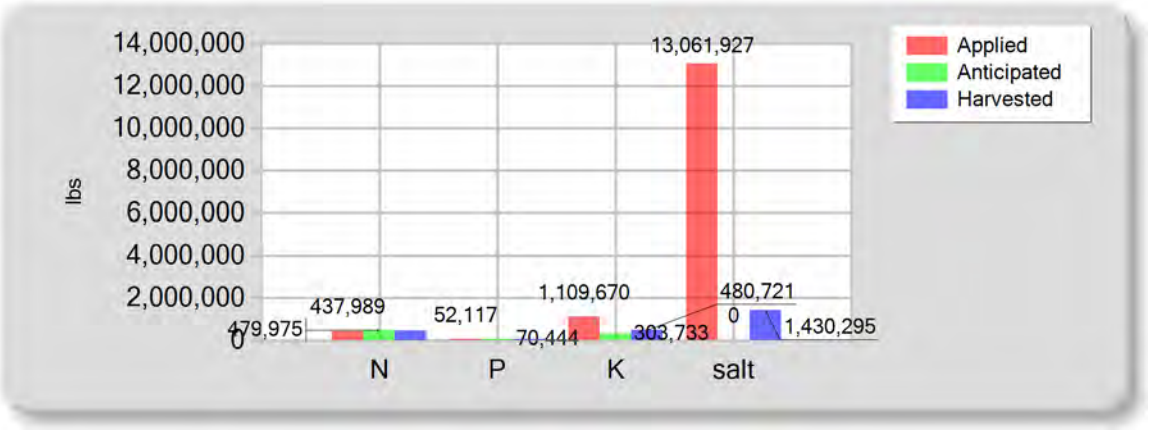
NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 18,450.00 | 18,450.00 | 18,450.00 | 0.00 |
| Dry manure | 78,086.54 | 9,177.93 | 26,014.25 | 0.00 |
| Process wastewater | 226,958.60 | 24,489.49 | 1,065,205.80 | 4,622,093.88 |
| Fresh water | 101,866.23 | 0.00 | 0.00 | 8,439,832.96 |
| Atmospheric deposition | 12,628.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 437,989.37 | 52,117.42 | 1,109,670.05 | 13,061,926.85 |
| Anticipated crop nutrient removal | 479,975.00 | 70,444.30 | 303,733.30 | 0.00 |
| Actual crop nutrient removal | 441,685.60 | 62,094.80 | 480,721.20 | 1,430,294.75 |
| Nutrient balance | -3,696.23 | -9,977.38 | 628,948.85 | 11,631,632.10 |
| Applied to removed ratio | 0.99 | 0.84 | 2.31 | 9.13 |

## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL



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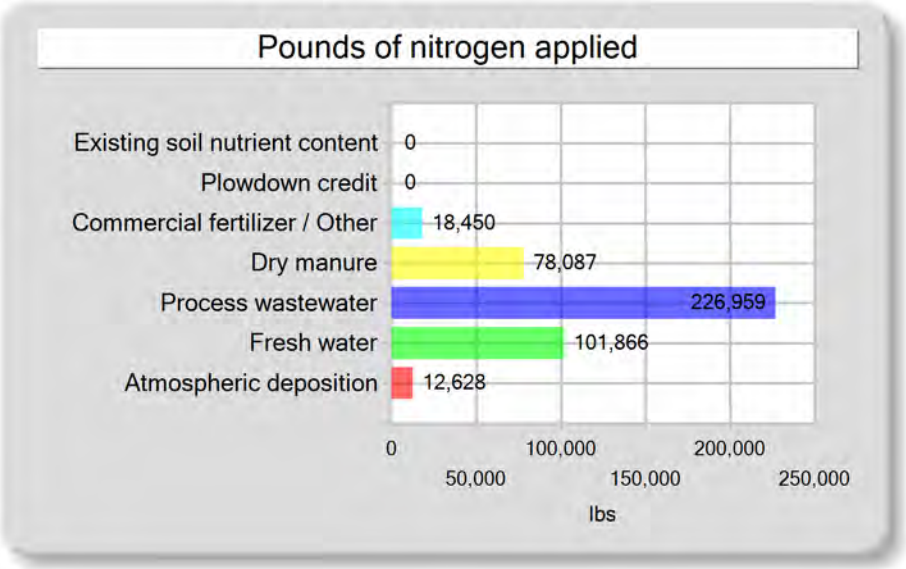
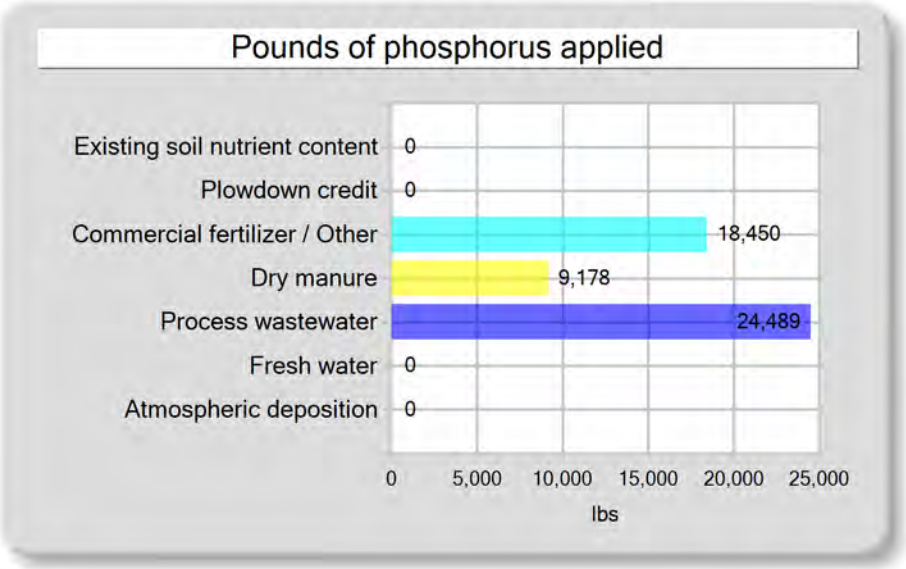
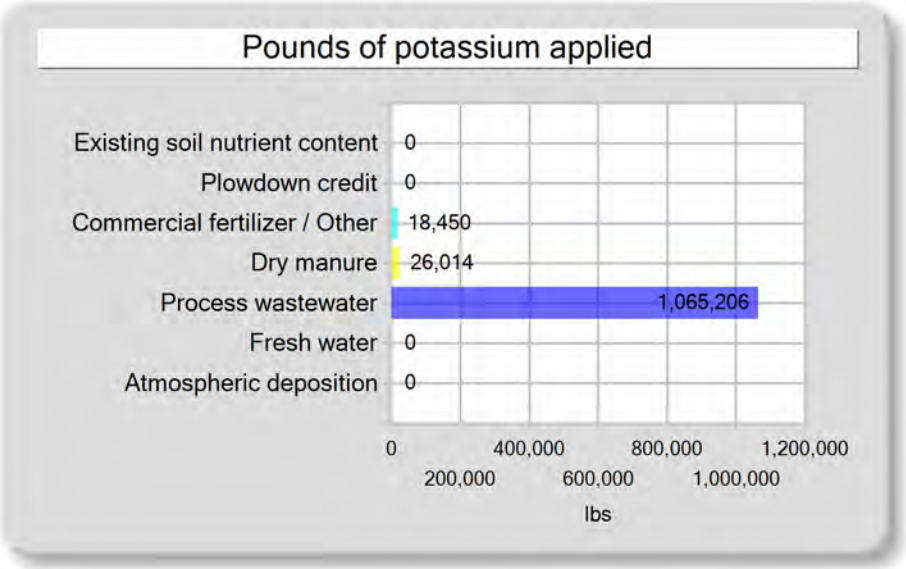
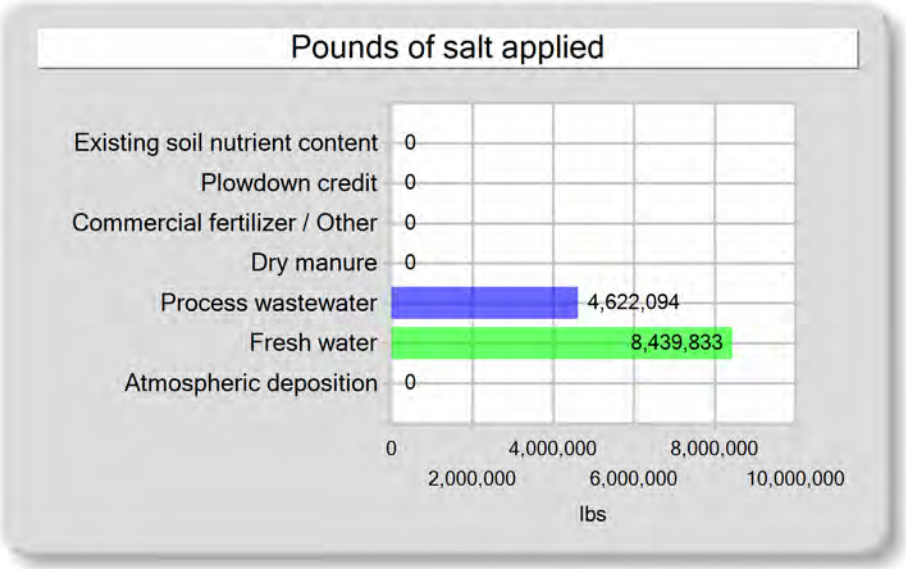
**C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE**



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## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | No |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

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ADDITIONAL NOTES

**A. NOTES**

Precipitation utilized during winter months to meet forage freshwater requirements. Heavy rains during the winter season allowed for smaller amounts of water to be applied through irrigation wells to grow crops.

Field 1 Corn had a lower than anticipated removal rate due to lower than anticipated %N and low tonnage. This resulted in field ratios slightly exceeding target limits.

Fields 3 Rye Grass, 3 Alfalfa, & 14 Corn had lower than anticipated removal rates. This was due to lower than expected yields and/or a lower than expected %N. The %N was based on analysis that was derived through a certified laboratory. However, the applications to these fields matched the low removal rates and was able to meet the field ratio threshold of 1.4.

Field 11 Rye Grass had a higher than anticipated removal rate. This was due to higher than expected %N. The %N removed, which is based on analysis derived through a certified laboratory, exceeded the anticipated values for this crop. Appropriate management practices resulted in a decent yield and a higher than expected nutrient removal rate.

Field 8 Alfalfa received no wastewater or solid manure in 2023. All nutrients applied to this field was contributed through freshwater applications and commercial fertilizer only.

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CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| DocuSigned by: 2E82D00B87114E0... | DocuSigned by: 1E10D6CE579949B... |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Frank Homen | Tony Homen |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 6/27/2024 | 6/26/2024 |
| DATE | DATE |

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ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

## OPERATOR INFORMATION

Name of Operator: Tony Homen

Name of Dairy Facility: Homen Dairy Farms

Facility Address:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 5511 W Sandy Mush RD | Merced | Merced | 95341-8404 |

Contact Person Name and Phone Number:

| Name | Phone Number |
|---|---|
| Tony Homen | (209) 564-1427 |

## MANURE HAULER INFORMATION

Name of Hauling Company/Person: Kenneth R Stones Spreading Services

Address of Hauling Company/Person:

| Number and Street | City | State | Zip Code |
|---|---|---|---|
| 5175 Shaw AVE | Winton | CA | 95388 |

Contact Person:

| Name | Phone Number |
|---|---|
| Paul Stone | (209) 756-1491 |

## DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

| Name | Phone Number |
|---|---|
| Stone Family Spreading | (209) 756-1491 |

| Address | City | State | Zip Code |
|---|---|---|---|
| 5175 W Shaw AVE | Winton | CA | 95388 |

Destination Address or Assessor's Parcel Number:

| Address | City | Zip Code |
|---|---|---|
| 5175 W Shaw AVE | Winton | 95388 |

| Street and nearest cross street (if no address) | County |
|---|---|
| | Merced |

| Assessor's Parcel Number | Assessor's Parcel Number County |
|---|---|
| | |

Last date hauled: 04/13/2023

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 7,799.26 tons

Manure Solids Content: 82.0 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

## CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
1E10D6CE579949B...
Operators Signature — Date: 6/26/2024

DocuSigned by:
1E10D6CE579949B...
Hauler Signature — Date: 6/26/2024

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

## OPERATOR INFORMATION

Name of Operator: Tony Homen

Name of Dairy Facility: Homen Dairy Farms

Facility Address:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 5511 W Sandy Mush RD | Merced | Merced | 95341-8404 |

Contact Person Name and Phone Number:

| Name | Phone Number |
|---|---|
| Tony Homen | (209) 564-1427 |

## MANURE HAULER INFORMATION

Name of Hauling Company/Person: Kenneth R Stones Spreading Services

Address of Hauling Company/Person:

| Number and Street | City | State | Zip Code |
|---|---|---|---|
| 5175 Shaw AVE | Winton | CA | 95388 |

Contact Person:

| Name | Phone Number |
|---|---|
| Paul Stone | (209) 756-1491 |

## DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

| Name | Phone Number |
|---|---|
| Stone Family Spreading | (209) 756-1491 |

| Address | City | State | Zip Code |
|---|---|---|---|
| 5175 W Shaw AVE | Winton | CA | 95388 |

Destination Address or Assessor's Parcel Number:

| Address | City | Zip Code |
|---|---|---|
| 5175 W Shaw AVE | Winton | 95388 |

| Street and nearest cross street (if no address) | County |
|---|---|
| | Merced |

| Assessor's Parcel Number | Assessor's Parcel Number County |
|---|---|
| | |

Last date hauled: 06/15/2023

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 2,827.33 tons

Manure Solids Content: 82.0 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

## CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
1E10D6CE579949B...
6/26/2024
Operator Signature — Date

DocuSigned by:
1E10D6CE579949B...
6/26/2024
Hauler Signature — Date

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

## OPERATOR INFORMATION

Name of Operator: Tony Homen

Name of Dairy Facility: Homen Dairy Farms

Facility Address:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 5511 W Sandy Mush RD | Merced | Merced | 95341-8404 |

Contact Person Name and Phone Number:

| Name | Phone Number |
|---|---|
| Tony Homen | (209) 564-1427 |

## MANURE HAULER INFORMATION

Name of Hauling Company/Person: Kenneth R Stones Spreading Services

Address of Hauling Company/Person:

| Number and Street | City | State | Zip Code |
|---|---|---|---|
| 5175 Shaw AVE | Winton | CA | 95388 |

Contact Person:

| Name | Phone Number |
|---|---|
| Paul Stone | (209) 756-1491 |

## DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

| Name | Phone Number |
|---|---|
| Flynn Ranch | (209) 564-1427 |

| Address | City | State | Zip Code |
|---|---|---|---|
| 8522 W Sandy Mush RD | Merced | CA | 95341 |

Destination Address or Assessor's Parcel Number:

| Address | City | Zip Code |
|---|---|---|
| 8522 W Sandy Mush RD | Merced | 95341 |

| Street and nearest cross street (if no address) | County |
|---|---|
| | Merced |

| Assessor's Parcel Number | Assessor's Parcel Number County |
|---|---|
| | |

Last date hauled: 09/15/2023

**Manure / Process Wastewater Tracking Manifest**
**For**
**Existing Milk Cow Dairies**

General Order No. R5-2007-0035, Attachment D

## MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,600.00 tons

Manure Solids Content: 26.4 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

## CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
1E10D6CE579949B...
Operator Signature — Date: 6/26/2024

DocuSigned by:
1E10D6CE579949B...
Hauler Signature — Date: 6/26/2024

DocuSign Envelope ID: C2B08CD0-FADC-4D69-AD9F-7F566095BC6C

DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 24B0718-01 | IW 1 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:00 |
| 24B0718-02 | IW 2 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:05 |
| 24B0718-03 | IW 3 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:10 |
| 24B0718-04 | IW 4 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:15 |
| 24B0718-05 | IW 5 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:18 |
| 24B0718-06 | IW 6 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:27 |
| 24B0718-07 | IW 7 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:35 |
| 24B0718-08 | IW 8 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:45 |
| 24B0718-09 | IW 9 | Ag Water | FR Ag Services | Freshwater - Irrigation Wells | 02/23/2024 6:55 |

Default Cooler Temperature on Receipt °C: 3.4

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results

**Sample: IW 1**
**24B0718-01 (Water)**

Sampled: 2/23/2024 6:00
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.83** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:25 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1830** | umhos/cm | 10.0 | 1 | | 02/26/24 12:25 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **15.1** | mg/L | 0.1 | 1 | 10 | 02/23/24 16:48 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:25 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.7** | units | 1.0 | 1 | | 02/26/24 12:25 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1120** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:52 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **15.1** | mg/L | 1.00 | 1 | | 02/27/24 11:52 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 2**
**24B0718-02 (Water)**

Sampled: 2/23/2024 6:05
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.83** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:27 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1830** | umhos/cm | 10.0 | 1 | | 02/26/24 12:27 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **14.9** | mg/L | 0.1 | 1 | 10 | 02/23/24 17:07 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:27 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.6** | units | 1.0 | 1 | | 02/26/24 12:27 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1020** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:54 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **14.9** | mg/L | 1.00 | 1 | | 02/27/24 11:54 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 3**
**24B0718-03 (Water)**

Sampled: 2/23/2024 6:10
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.82** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:28 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **822** | umhos/cm | 10.0 | 1 | | 02/26/24 12:28 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **5.5** | mg/L | 0.1 | 1 | 10 | 02/23/24 17:27 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:28 | SM 4500-H+ | H | BFB0552 |
| **pH** | **8.0** | units | 1.0 | 1 | | 02/26/24 12:28 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **440** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:55 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **5.49** | mg/L | 1.00 | 1 | | 02/27/24 11:55 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By:

Received: 12/15/2022 15:20
Reported: 01/27/2023 08:39

## Sample Results
**(Continued)**

**Sample: IW 4**
**22L0902-04 (Water)**

Sampled: 12/15/2022 9:48
Sampled By: Tony

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.51** | mmhos/cm | 0.01 | 1 | | 12/19/22 10:15 | SM 2510 B | | BDL0618 |
| **Electrical Conductivity umhos** | **1510** | umhos/cm | 10.0 | 1 | | 12/19/22 10:15 | SM 2510 B | | BDL0618 |
| **Nitrate Nitrogen as NO3N** | **1.8** | mg/L | 0.1 | 1 | 10 | 12/15/22 22:53 | EPA 300.0 | | BDL0588 |
| **pH** | **7.8** | units | 1.0 | 1 | | 12/19/22 10:15 | SM 4500-H+ | H | BDL0618 |
| **Total Filterable Solids (TDS)** | **783** | mg/L | 10.0 | 1 | | 12/29/22 12:48 | SM 2540 C | | BDL0635 |
| **Temperature** | **25.0** | °C | 0.0 | 1 | | 12/19/22 10:15 | SM 2510 B | | BDL0618 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 01/04/23 10:19 | SM 4500-NH3 C | | BDL0748 |
| **Total Nitrogen** | **1.96** | mg/L | 1.00 | 1 | | 01/04/23 10:19 | SM 4500-NH3 C | | BDL0748 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 4**
**24B0718-04 (Water)**

Sampled: 2/23/2024 6:15
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **3.91** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:30 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **3910** | umhos/cm | 10.0 | 1 | | 02/26/24 12:30 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **58.0** | mg/L | 0.1 | 1 | 10 | 02/23/24 17:47 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:30 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.8** | units | 1.0 | 1 | | 02/26/24 12:30 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **2280** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:56 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **58.9** | mg/L | 1.00 | 1 | | 02/27/24 11:56 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 5**
**24B0718-05 (Water)**

Sampled: 2/23/2024 6:18
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.84** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:31 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1840** | umhos/cm | 10.0 | 1 | | 02/26/24 12:31 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **15.0** | mg/L | 0.1 | 1 | 10 | 02/23/24 18:06 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:31 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.8** | units | 1.0 | 1 | | 02/26/24 12:31 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1130** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:58 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **15.1** | mg/L | 1.00 | 1 | | 02/27/24 11:58 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 6**
**24B0718-06 (Water)**

Sampled: 2/23/2024 6:27
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.79** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:39 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1790** | umhos/cm | 10.0 | 1 | | 02/26/24 12:39 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **14.8** | mg/L | 0.1 | 1 | 10 | 02/23/24 18:26 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:39 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.6** | units | 1.0 | 1 | | 02/26/24 12:39 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1130** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 11:59 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **15.2** | mg/L | 1.00 | 1 | | 02/27/24 11:59 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 7**
**24B0718-07 (Water)**

Sampled: 2/23/2024 6:35
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.82** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:41 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1820** | umhos/cm | 10.0 | 1 | | 02/26/24 12:41 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **14.8** | mg/L | 0.1 | 1 | 10 | 02/23/24 18:46 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:41 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.8** | units | 1.0 | 1 | | 02/26/24 12:41 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1110** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 12:00 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **14.8** | mg/L | 1.00 | 1 | | 02/27/24 12:00 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 8**
**24B0718-08 (Water)**

Sampled: 2/23/2024 6:45
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.82** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:43 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **817** | umhos/cm | 10.0 | 1 | | 02/26/24 12:43 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **5.5** | mg/L | 0.1 | 1 | 10 | 02/23/24 19:05 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:43 | SM 4500-H+ | H | BFB0552 |
| **pH** | **8.1** | units | 1.0 | 1 | | 02/26/24 12:43 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **490** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 12:02 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **5.50** | mg/L | 1.00 | 1 | | 02/27/24 12:02 | SM 4500-NH3 C | | BFB0536 |

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DELLAVALLE™ LABORATORY INC

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7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Sample Results
**(Continued)**

**Sample: IW 9**
**24B0718-09 (Water)**

Sampled: 2/23/2024 6:55
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.84** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:44 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1840** | umhos/cm | 10.0 | 1 | | 02/26/24 12:44 | SM 2510 B | | BFB0552 |
| **Nitrate Nitrogen as NO3N** | **15.0** | mg/L | 0.1 | 1 | 10 | 02/23/24 19:25 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:44 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.7** | units | 1.0 | 1 | | 02/26/24 12:44 | SM 4500-H+ | H | BFB0552 |
| **Total Filterable Solids (TDS)** | **1120** | mg/L | 10.0 | 1 | | 02/27/24 14:26 | SM 2540 C | | BFB0520 |
| Kjeldahl Nitrogen (TKN), Total | ND | mg/L | 1.00 | 1 | | 02/27/24 12:03 | SM 4500-NH3 C | | BFB0536 |
| **Total Nitrogen** | **15.0** | mg/L | 1.00 | 1 | | 02/27/24 12:03 | SM 4500-NH3 C | | BFB0536 |

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Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0520*** | | | | | | | | | | |
| **Blank (BFB0520-BLK1)** | | | | | Prepared: 2/23/2024 | Analyzed: 2/27/2024 | | | | |
| Total Filterable Solids (TDS) | ND | | 10.0 | mg/L | | | | | | |
| **LCS (BFB0520-BS1)** | | | | | Prepared: 2/23/2024 | Analyzed: 2/27/2024 | | | | |
| Total Filterable Solids (TDS) | 26.2 | | 10.0 | mg/L | 2000 | | 1.31 | 0-200 | | |
| **Duplicate (BFB0520-DUP1)** | | **Source: 24B0668-01** | | | Prepared: 2/23/2024 | Analyzed: 2/27/2024 | | | | |
| Total Filterable Solids (TDS) | 880 | | 10.0 | mg/L | | 890 | | | 1.13 | 10 |
| **Duplicate (BFB0520-DUP2)** | | **Source: 24B0718-08** | | | Prepared: 2/23/2024 | Analyzed: 2/27/2024 | | | | |
| Total Filterable Solids (TDS) | 480 | | 10.0 | mg/L | | 490 | | | 2.06 | 10 |
| **Reference (BFB0520-SRM1)** | | | | | Prepared: 2/23/2024 | Analyzed: 2/27/2024 | | | | |
| Total Filterable Solids (TDS) | 390 | | | mg/L | 390.0 | | 100 | 90-110 | | |

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7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0526*** | | | | | | | | | | |
| **Blank (BFB0526-BLK1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BFB0526-BLK2)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BFB0526-BLK3)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BFB0526-BS1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.4 | | 0.1 | mg/L | 5.000 | | 108 | 90-110 | | |
| **LCS (BFB0526-BS2)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.3 | | 0.1 | mg/L | 5.000 | | 106 | 90-110 | | |
| **Duplicate (BFB0526-DUP1)** | | **Source: 24B0718-03** | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.5 | | 0.1 | mg/L | | 5.5 | | | 0.255 | 10 |
| **Duplicate (BFB0526-DUP2)** | | **Source: 24B0719-03** | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.5 | | 0.1 | mg/L | | 5.5 | | | 0.492 | 10 |
| **Matrix Spike (BFB0526-MS1)** | | **Source: 24B0718-03** | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.6 | | 0.1 | mg/L | 5.000 | 5.5 | 102 | 90-110 | | |
| **Matrix Spike (BFB0526-MS2)** | | **Source: 24B0719-03** | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.8 | | 0.1 | mg/L | 5.000 | 5.5 | 106 | 90-110 | | |
| **Reference (BFB0526-SRM1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BFB0526-SRM2)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BFB0526-SRM3)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0536*** | | | | | | | | | | |
| **Blank (BFB0536-BLK1)** | | | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.00 | mg/L | | | | | | |
| Total Nitrogen | ND | | 1.00 | mg/L | | | | | | |
| **Blank (BFB0536-BLK2)** | | | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.00 | mg/L | | | | | | |
| Total Nitrogen | ND | | 1.00 | mg/L | | | | | | |
| **LCS (BFB0536-BS1)** | | | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 2.96 | | 1.00 | mg/L | 2.854 | | 104 | 90-110 | | |
| **LCS (BFB0536-BS2)** | | | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 3.13 | | 1.00 | mg/L | 2.854 | | 110 | 90-110 | | |
| **Duplicate (BFB0536-DUP1)** | | **Source: 24B0718-03** | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | ND | | 1.40 | mg/L | | ND | | | | 10 |
| **Duplicate (BFB0536-DUP2)** | | **Source: 24B0728-01** | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 27.7 | | 3.50 | mg/L | | 27.4 | | | 0.961 | 10 |
| **Matrix Spike (BFB0536-MS1)** | | **Source: 24B0718-03** | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 4.34 | | 1.40 | mg/L | 7.992 | ND | 54.3 | 90-110 | | |
| **Matrix Spike (BFB0536-MS2)** | | **Source: 24B0728-01** | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 36.4 | | 3.50 | mg/L | 9.990 | 27.4 | 89.7 | 90-110 | | |
| **Reference (BFB0536-SRM1)** | | | | | Prepared: 2/26/2024 | Analyzed: 2/27/2024 | | | | |
| Kjeldahl Nitrogen (TKN), Total | 34.8 | | | mg/L | 31.90 | | 109 | 90-110 | | |

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Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0552*** | | | | | | | | | | |
| **Blank (BFB0552-BLK1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.5 | | 1.0 | units | | | | | | |
| **Blank (BFB0552-BLK2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 7.8 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BFB0552-BLK3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 8.0 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Duplicate (BFB0552-DUP1)** | | **Source: 24B0710-03** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 5.20 | | 0.01 | mmhos/cm | | 5.16 | | | 0.869 | 10 |
| pH | 7.1 | | 1.0 | units | | 7.0 | | | 0.567 | 1 |
| Electrical Conductivity umhos | 5200 | | 10.0 | umhos/cm | | 5160 | | | 0.869 | 10 |
| **Duplicate (BFB0552-DUP2)** | | **Source: 24B0719-01** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1.81 | | 0.01 | mmhos/cm | | 1.80 | | | 0.471 | 10 |
| Electrical Conductivity umhos | 1810 | | 10.0 | umhos/cm | | 1800 | | | 0.471 | 10 |
| pH | 7.7 | | 1.0 | units | | 7.7 | | | 0.259 | 1 |
| **Reference (BFB0552-SRM1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 445 | | | umhos/cm | 426.0 | | 105 | 90-110 | | |
| **Reference (BFB0552-SRM2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 67021-101.32 | | |
| **Reference (BFB0552-SRM3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| Electrical Conductivity umhos | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| **Reference (BFB0552-SRM4)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| Electrical Conductivity umhos | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| **Reference (BFB0552-SRM5)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |

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7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:19

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0552 (Continued)*** | | | | | | | | | | |
| **Reference (BFB0552-SRM5)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity umhos | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |
| **Reference (BFB0552-SRM6)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0552-SRM7)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0552-SRM8)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |

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02/23/24 10:35 24B0718 JG

# DELLAVALLE LABORATOR

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 9372
www.dellavallelab.com 559 233-6129 · 800 228-9896 · F

No. Samples: 9 No of Bottles:

Purchase Order No

Bill To: 22210 (Acct #) | 8 (Cons #)

Results Need By

**Water Type:** [ ] Drinking Water [ ] W
[✓] Ag Water [ ] Groundwater [ ] M

**Other:**

**Name:** Homen Dairy Farms
**Address:** 7496 Combs Road
**City:** Merced **State:** CA **Zip:** 95341
**Telephone:** **Fax:**
**Cell/Email:** homendairyfarms@aol.com
**COPY TO:** labs@fragservices.com
krhodes@fragservices.com ariordan@fragservices.com
**REQUESTED BY:** FR Ag
**PROJECT:**
**CROP:** FRESHWATER IRRIGATION WELLS

[X] Copy of Chain [X] QA/QC Documents
Sampled By: FR Ag Services

**Analysis and Bottles Required:** *(Please indicate*

( ) DWW1: EC, $NO_3$-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca,
(1-1 Liter Plastic, Unpreserved) White Per Sam
(✓) DCW1: EC, $NO_3$-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DPW1: EC, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DPW2: DPW1 Plus Ca, Mg, Na, $HCO_3$, $CO_3$, S
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C |
|---|---|---|---|---|
| 1 | IW 1 | 2/23 | 0600 | 3.4 |
| 2 | 2 | | 0605 | 0.5 |
| 3 | 3 | | 0610 | -0.5 |
| 4 | 4 | | 0615 | -1.3 |
| 5 | 5 | | 0618 | -1.1 |
| 6 | 6 | | 0623 | 1.9 |
| 7 | 7 | | 0635 | 2.9 |
| 8 | 8 | | 0645 | 1.1 |
| 9 | 9 | | 0655 | -0.2 |
| 10 | | | | |

IR Thermometer SN: 221511276
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Hanford

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | Kierstan Rhodes | FR Ag Services | 2/22 0455 | |
| Second | [illegible] | DLI | 2/23 7:55 | 2/23 944 |
| Third | | | | |
| Fourth | [illegible] | DLI | 2/23/24 10:35 | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

***Invoicing Information:***
Sampling hrs
Miles
Consulting

***Shipping***
$ In
$ Out

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Sample received in cooler with ice (cool
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[Homen Dairy Farms Water.xls]Dairy Water

DocuSign Envelope ID: C2B08CD0-FADC-4D69-AD9F-7F566095BC6C

02/23/24 10:35 24B0718

**Shipping Information:** Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

**Container:** Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

**Samples Preserved with $HNO_3$ or $H_2SO_4$ were:** ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

| | Type of Container(s) Received | Sample Number 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Sample Containers for Internal (DLI) Use** *(Containers that go into the Lab)* | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Plastics | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | * pH Value | | | | | | | | | | |
| Plastics | 250 mL $H_2SO_4$ (Yellow) Plastic | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | |
| Plastics | * pH Value | <2 | <2 | <2 | <2 | <2 | <2 | <2 | <2 | <2 | |
| Plastics | 500 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (White) Plastic | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | |
| Plastics | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| Special | 1 L AG unpreserved (White) | | | | | | | | | | |
| Special | PO4-P Kit | | | | | | | | | | |
| Special | Other: | | | | | | | | | | |
| | **Sample Containers for Subcontracted ("Send Out") Analyses** *(Containers that go in the Subcontract ("Send Out") Refrigerator)* | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Plastics | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| Plastics | 1 L $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | 500 mL Amber Plastic HCl (Blue) (Ferrous Iron) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ (EPA547) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA HCl + Asc. Acid (Blue)(Set of 2) (123 TCP) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3) (TTHM) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $H_3PO_4$ (Set of 3) (TOC) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, HCl (Blue) (Set of 3) (EPA524) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, HCl (Blue) (Set of 3) (123 TCP) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3) (EPA504) | | | | | | | | | | |
| Glass | 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| Glass | 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Glass | 250 mL AG Ammonium Chlorite (HAA5) | | | | | | | | | | |
| Glass | 500 mL glass unpreserved (White) | | | | | | | | | | |
| Glass | 1 L AG unpreserved (White) | | | | | | | | | | |
| Glass | 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| Glass | 1 L AG $Na_2S_2O_3$ (Green) (EPA 549) | | | | | | | | | | |
| Glass | 1 L AG HCl (Blue) (Oil & Grease) | | | | | | | | | | |
| Glass | 1 L AG Asc. Acid, EDTA, $KH_2CH$ (Set of 2) (EPA 507/525) | | | | | | | | | | |
| Special | $Cr^{6+}$ - 250 mL Plastic w/$HN_4ON$ (N) $SO_4$ | | | | | | | | | | |
| Special | Cyanide - 250 mL NaOH | | | | | | | | | | |
| Special | Asbestos - 1 L Plastic wrapped in foil (Set of 2) | | | | | | | | | | |
| Special | Sulfide - 1 L Plastic NaOH + ZnAc | | | | | | | | | | |
| Special | Chlorite/Bromate - 250 mL AG with EDA | | | | | | | | | | |
| Special | DO KIT | | | | | | | | | | |
| Special | Other: | | | | | | | | | | |
| Special | Other: | | | | | | | | | | |

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SEND OUT

DocuSign Envelope ID: C2B08CD0-FADC-4D69-AD9F-7F566095BC6C

DELLAVALLE™
LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 24B0719-01 | Dom 1E | Drinking Water | FR Ag Services | Freshwater - Dom | 02/23/2024 0:00 |
| 24B0719-02 | Dom 2W | Drinking Water | FR Ag Services | Freshwater - Dom | 02/23/2024 6:45 |
| 24B0719-03 | Dom 3 Heifer | Drinking Water | FR Ag Services | Freshwater - Dom | 02/23/2024 6:52 |
| 24B0719-04 | Dom 4 House | Drinking Water | FR Ag Services | Freshwater - Dom | 02/23/2024 6:58 |
| 24B0719-05 | Dom 5 Heifers | Drinking Water | FR Ag Services | Freshwater - Dom | 02/23/2024 7:15 |

Default Cooler Temperature on Receipt °C: 2.6
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Sample Results

**Sample: Dom 1E**
**24B0719-01 (Water)**

Sampled: 2/23/2024 0:00
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.80** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:46 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1800** | umhos/cm | 10.0 | 1 | | 02/26/24 12:46 | SM 2510 B | | BFB0552 |
| Ammonia (as N) | * | mg/L | 0.00 | 1 | | 02/28/24 12:28 | Field | | BFB0618 |
| **Nitrate Nitrogen as NO3N** | **14.7** | mg/L | 0.1 | 1 | 10 | 02/23/24 19:45 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:46 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.7** | units | 1.0 | 1 | | 02/26/24 12:46 | SM 4500-H+ | H | BFB0552 |

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## Sample Results
**(Continued)**

**Sample: Dom 2W**
**24B0719-02 (Water)**

Sampled: 2/23/2024 6:45
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **1.79** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:47 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **1790** | umhos/cm | 10.0 | 1 | | 02/26/24 12:47 | SM 2510 B | | BFB0552 |
| Ammonia (as N) | * | mg/L | 0.00 | 1 | | 02/28/24 12:28 | Field | | BFB0618 |
| **Nitrate Nitrogen as NO3N** | **14.9** | mg/L | 0.1 | 1 | 10 | 02/23/24 22:23 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:47 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.8** | units | 1.0 | 1 | | 02/26/24 12:47 | SM 4500-H+ | H | BFB0552 |

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Merced, CA 95341

Account# 00-0022210
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Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Sample Results
**(Continued)**

**Sample: Dom 3 Heifer**
**24B0719-03 (Water)**

Sampled: 2/23/2024 6:52
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **0.82** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:49 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **824** | umhos/cm | 10.0 | 1 | | 02/26/24 12:49 | SM 2510 B | | BFB0552 |
| Ammonia (as N) | * | mg/L | 0.00 | 1 | | 02/28/24 12:28 | Field | | BFB0618 |
| **Nitrate Nitrogen as NO3N** | **5.5** | mg/L | 0.1 | 1 | 10 | 02/23/24 22:43 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:49 | SM 4500-H+ | H | BFB0552 |
| **pH** | **8.0** | units | 1.0 | 1 | | 02/26/24 12:49 | SM 4500-H+ | H | BFB0552 |

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Received: 02/23/2024 10:35
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## Sample Results
**(Continued)**

**Sample: Dom 4 House**
**24B0719-04 (Water)**

Sampled: 2/23/2024 6:58
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **3.73** | mmhos/cm | 0.01 | 1 | | 02/26/24 14:41 | SM 2510 B | | BFB0554 |
| **Electrical Conductivity umhos** | **3730** | umhos/cm | 10.0 | 1 | | 02/26/24 14:41 | SM 2510 B | | BFB0554 |
| Ammonia (as N) | * | mg/L | 0.00 | 1 | | 02/28/24 12:28 | Field | | BFB0618 |
| **Nitrate Nitrogen as NO3N** | **56.5** | mg/L | 0.1 | 1 | 10 | 02/23/24 23:02 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 14:41 | SM 4500-H+ | H | BFB0554 |
| **pH** | **7.5** | units | 1.0 | 1 | | 02/26/24 14:41 | SM 4500-H+ | H | BFB0554 |

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Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Sample Results
**(Continued)**

**Sample: Dom 5 Heifers**
**24B0719-05 (Water)**

Sampled: 2/23/2024 7:15
Sampled By: FR Ag Services

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **3.93** | mmhos/cm | 0.01 | 1 | | 02/26/24 12:50 | SM 2510 B | | BFB0552 |
| **Electrical Conductivity umhos** | **3930** | umhos/cm | 10.0 | 1 | | 02/26/24 12:50 | SM 2510 B | | BFB0552 |
| Ammonia (as N) | * | mg/L | 0.00 | 1 | | 02/28/24 12:28 | Field | | BFB0618 |
| **Nitrate Nitrogen as NO3N** | **56.3** | mg/L | 0.1 | 1 | 10 | 02/23/24 23:22 | EPA 300.0 | | BFB0526 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 02/26/24 12:50 | SM 4500-H+ | H | BFB0552 |
| **pH** | **7.8** | units | 1.0 | 1 | | 02/26/24 12:50 | SM 4500-H+ | H | BFB0552 |

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Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0526*** | | | | | | | | | | |
| **Blank (BFB0526-BLK1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BFB0526-BLK2)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BFB0526-BLK3)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BFB0526-BS1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.4 | | 0.1 | mg/L | 5.000 | | 108 | 90-110 | | |
| **LCS (BFB0526-BS2)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.3 | | 0.1 | mg/L | 5.000 | | 106 | 90-110 | | |
| **Duplicate (BFB0526-DUP1)** | | **Source: 24B0718-03** | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.5 | | 0.1 | mg/L | | 5.5 | | | 0.255 | 10 |
| **Duplicate (BFB0526-DUP2)** | | **Source: 24B0719-03** | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 5.5 | | 0.1 | mg/L | | 5.5 | | | 0.492 | 10 |
| **Matrix Spike (BFB0526-MS1)** | | **Source: 24B0718-03** | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.6 | | 0.1 | mg/L | 5.000 | 5.5 | 102 | 90-110 | | |
| **Matrix Spike (BFB0526-MS2)** | | **Source: 24B0719-03** | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.8 | | 0.1 | mg/L | 5.000 | 5.5 | 106 | 90-110 | | |
| **Reference (BFB0526-SRM1)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BFB0526-SRM2)** | | | | | Prepared & Analyzed: 2/23/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |
| **Reference (BFB0526-SRM3)** | | | | | Prepared: 2/23/2024 Analyzed: 2/24/2024 | | | | | |
| Nitrate Nitrogen as NO3N | 10.3 | | | mg/L | 10.00 | | 103 | 90-110 | | |

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Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

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Reported: 02/28/2024 14:14

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0552*** | | | | | | | | | | |
| **Blank (BFB0552-BLK1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 6.5 | | 1.0 | units | | | | | | |
| **Blank (BFB0552-BLK2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.8 | | 1.0 | units | | | | | | |
| **Blank (BFB0552-BLK3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 8.0 | | 1.0 | units | | | | | | |
| **Duplicate (BFB0552-DUP1)** | | **Source: 24B0710-03** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 5.20 | | 0.01 | mmhos/cm | | 5.16 | | | 0.869 | 10 |
| Electrical Conductivity umhos | 5200 | | 10.0 | umhos/cm | | 5160 | | | 0.869 | 10 |
| pH | 7.1 | | 1.0 | units | | 7.0 | | | 0.567 | 1 |
| **Duplicate (BFB0552-DUP2)** | | **Source: 24B0719-01** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1.81 | | 0.01 | mmhos/cm | | 1.80 | | | 0.471 | 10 |
| Electrical Conductivity umhos | 1810 | | 10.0 | umhos/cm | | 1800 | | | 0.471 | 10 |
| pH | 7.7 | | 1.0 | units | | 7.7 | | | 0.259 | 1 |
| **Reference (BFB0552-SRM1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 445 | | | umhos/cm | 426.0 | | 105 | 90-110 | | |
| **Reference (BFB0552-SRM2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 67021-101.32 | | |
| **Reference (BFB0552-SRM3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| Electrical Conductivity umhos | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| **Reference (BFB0552-SRM4)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| Electrical Conductivity umhos | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| **Reference (BFB0552-SRM5)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |

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## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0552 (Continued)*** | | | | | | | | | | |
| **Reference (BFB0552-SRM5)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity umhos | 1090 | | | umhos/cm | 1000 | | 109 | 90-110 | | |
| **Reference (BFB0552-SRM6)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0552-SRM7)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0552-SRM8)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0554*** | | | | | | | | | | |
| **Blank (BFB0554-BLK1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| pH | 5.8 | | 1.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| **Blank (BFB0554-BLK2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.5 | | 1.0 | units | | | | | | |
| **Blank (BFB0554-BLK3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 8.0 | | 1.0 | units | | | | | | |
| **Duplicate (BFB0554-DUP1)** | | **Source: 24B0722-05** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 0.22 | | 0.01 | mmhos/cm | | 0.24 | | | 5.09 | 10 |
| Electrical Conductivity umhos | 224 | | 10.0 | umhos/cm | | 236 | | | 5.09 | 10 |
| pH | 7.5 | | 1.0 | units | | 7.8 | | | 4.45 | 1 |
| **Duplicate (BFB0554-DUP2)** | | **Source: 24B0719-04** | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| Electrical Conductivity | 3.79 | | 0.01 | mmhos/cm | | 3.73 | | | 1.65 | 10 |
| Electrical Conductivity umhos | 3790 | | 10.0 | umhos/cm | | 3730 | | | 1.65 | 10 |
| pH | 7.8 | | 1.0 | units | | 7.5 | | | 3.92 | 1 |
| **Reference (BFB0554-SRM1)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 9.1 | | 1.0 | units | | | | 97.5-102.5 | | |
| **Reference (BFB0554-SRM2)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 7.6 | | | units | 7.520 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0554-SRM3)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 6.4 | | 1.0 | units | | | | 97.5-102.5 | | |
| **Reference (BFB0554-SRM4)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 6.4 | | 1.0 | units | | | | 97.5-102.5 | | |
| **Reference (BFB0554-SRM5)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 7.2 | | 1.0 | units | | | | 97.5-102.5 | | |
| **Reference (BFB0554-SRM6)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |

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DELLAVALLE™ LABORATORY INC

Homen Dairy Farms
7496 Combs Road
Merced, CA 95341

Account# 00-0022210
Account Manager: Ben Nydam
Submitted By: FR Ag

Received: 02/23/2024 10:35
Reported: 02/28/2024 14:14

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BFB0554 (Continued)*** | | | | | | | | | | |
| **Reference (BFB0554-SRM6)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0554-SRM7)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |
| **Reference (BFB0554-SRM8)** | | | | | Prepared & Analyzed: 2/26/2024 | | | | | |
| pH | 4.0 | | | units | 4.000 | | 101 | 97.5-102.5 | | |

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02/23/24 10:35 24B0719 JG

# DELLAVALLE LABORATOR

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 9372
www.dellavallelab.com 559 233-6129 · 800 228-9896 · F

No. Samples: 5 No of Bottles: 10

Purchase Order No

Bill To: 22210 Acct # | 8 Cons #

Results Need By

**Water Type:** [X] Drinking Water [ ] W
[ ] Ag Water [ ] Groundwater [ ] M

**Name:** Homen Dairy Farms
**Address:** 7496 Combs Road
**City:** Merced **State:** CA **Zip:** 95341
**Telephone:** **Fax:**
**Cell/Email:** homendairyfarms@aol.com
**COPY TO:** labs@fragservices.com
krhodes@fragservices.com ariordan@fragservices.com
**REQUESTED BY:** FR Ag
**PROJECT:**
**CROP:** Freshwater - Dom

[X] Copy of Chain [X] QA/QC Documents
Sampled By: FR Ag Services

**Other:**

**Analysis and Bottles Required: (*Please indicate*)**

(X) DWW1: EC, $NO_3$-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca,
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DCW1: EC, $NO_3$-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DPW1: EC, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) DPW2: DPW1 Plus Ca, Mg, Na, $HCO_3$, $CO_3$, S
(1-1 Liter Plastic, Unpreserved) White Per Sam
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C |
|---|---|---|---|---|
| 14.7 1 | Dom 1E | 2/23 | | 2.6 |
| 14.9 2 | Dom 2W | 0645 | | 3.8 |
| 5.5 3 | Dom 3 Heifer | 0652 | | -0.8 |
| 56.5 4 | Dom 4 House | 0658 | | -0.2 |
| 56 5 | Dom 5 Heifers | 0715 | | 2.0 |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |
| 9 | | | | |
| 10 | | | | |

100.0 (mg/L)

AWS 2/23/24

IR Thermometer SN: 221511276
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Hanford

Notes: Jamie - 2/26/24 14:02 - Per Kierstan @ FR Ag, OK & thanks

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | Kierstan Rhodes | FR Ag Services | 2/23 0715 | |
| Second | | DLI | 2/23 7:55 | 2/23 9:44 |
| Third | | | | |
| Fourth | | DLI | 2/23/24 10:35 | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

***Invoicing Information:***
Sampling hrs
Miles
Consulting

***Shipping***
$ In
$ Out

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Sample received in cooler with ice (cool
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[Homen Dairy Farms Water.xls]Dairy Water

DocuSign Envelope ID: C2B08CD0-FADC-4D69-AD9F-7F566095BC6C

02/23/24 10:35 24B0719

**Shipping Information:** Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

**Container:** Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

**Samples Preserved with $HNO_3$ or $H_2SO_4$ were:** ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

| | Type of Container(s) Received | Sample Number 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Sample Containers for Internal (DLI) Use** *(Containers that go into the Lab)* | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Plastics | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | * pH Value | | | | | | | | | | |
| Plastics | 250 mL $H_2SO_4$ (Yellow) Plastic | 1 | 1 | 1 | 1 | 1 | | | | | |
| Plastics | * pH Value | <2 | <2 | <2 | <2 | <2 | | | | | |
| Plastics | 500 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (White) Plastic | 1 | 1 | 1 | 1 | 1 | | | | | |
| Plastics | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| Special | 1 L AG unpreserved (White) | | | | | | | | | | |
| Special | PO4-P Kit | | | | | | | | | | |
| Special | **Other:** | | | | | | | | | | |
| | **Sample Containers for Subcontracted ("Send Out") Analyses** *(Containers that go in the Subcontract ("Send Out") Refrigerator)* | | | | | | | | | | |
| Plastics | 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Plastics | 250 mL unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 250 mL $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | 250 mL $H_2SO_4$ (Yellow) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (White) Plastic | | | | | | | | | | |
| Plastics | 1 L unpreserved (BOD) (Purple) Plastic | | | | | | | | | | |
| Plastics | 1 L $HNO_3$ (Red) Plastic | | | | | | | | | | |
| Plastics | 500 mL Amber Plastic HCl (Blue) (Ferrous Iron) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ (EPA547) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA HCl + Asc. Acid (Blue)(Set of 2) (123 TCP) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3) (TTHM) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $H_3PO_4$ (Set of 3) (TOC) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, HCl (Blue) (Set of 3) (EPA524) | | | | | | | | | | |
| VOA Vials | 40 mL AG VOA, HCl (Blue) (Set of 3) (123 TCP) | | | | | | | | | | |
| VOA Vials | 40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3) (EPA504) | | | | | | | | | | |
| Glass | 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| Glass | 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| Glass | 250 mL AG Ammonium Chlorite (HAA5) | | | | | | | | | | |
| Glass | 500 mL glass unpreserved (White) | | | | | | | | | | |
| Glass | 1 L AG unpreserved (White) | | | | | | | | | | |
| Glass | 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| Glass | 1 L AG $Na_2S_2O_3$ (Green) (EPA 549) | | | | | | | | | | |
| Glass | 1 L AG HCl (Blue) (Oil & Grease) | | | | | | | | | | |
| Glass | 1 L AG Asc. Acid, EDTA, $KH_2CH$ (Set of 2) (EPA 507/525) | | | | | | | | | | |
| Special | $Cr^{6+}$ - 250 mL Plastic w/$HN_4ON$ (N) $SO_4$ | | | | | | | | | | |
| Special | Cyanide - 250 mL NaOH | | | | | | | | | | |
| Special | Asbestos - 1 L Plastic wrapped in foil (Set of 2) | | | | | | | | | | |
| Special | Sulfide - 1 L Plastic NaOH + ZnAc | | | | | | | | | | |
| Special | Chlorite/Bromate - 250 mL AG with EDA | | | | | | | | | | |
| Special | DO KIT | | | | | | | | | | |
| Special | **Other:** | | | | | | | | | | |
| Special | **Other:** | | | | | | | | | | |

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SEND OUT